

Report of Project 5: Turbulence Modeling using FLUENT

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1 Center-line Velocity Comparison - Q2(i)

For a hydrogen-air flame, the center-line of the flame is an area of importance. In the fluent model, turbulence is accounted for to simulate the real world. The turbulence intensity is set to 2% and the length-scale to 0.002 m at the air inlet. At the air outlet, $k = 10 \frac{m^2}{s^2}$ and $\varepsilon = 200 \frac{m^2}{s^3}$. The initial conditions are $U = 1$, $k = 0.01$, $\varepsilon = 10$, and $\xi = 0$.

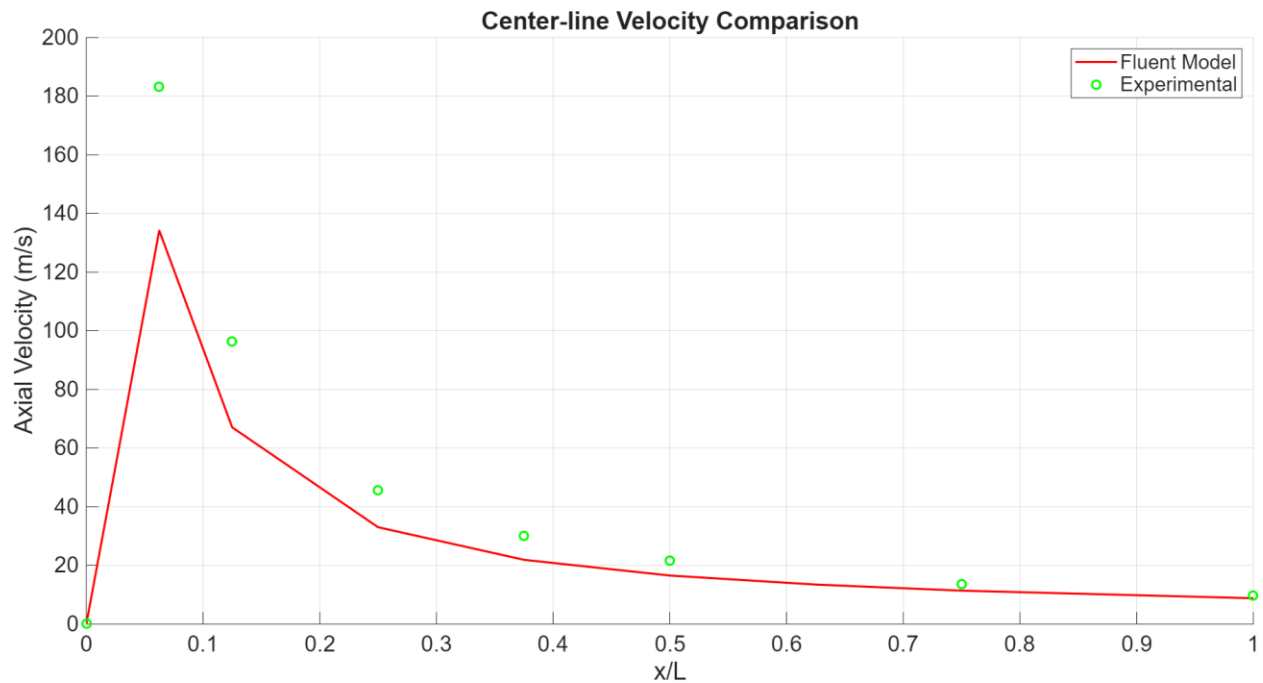


Figure 1 Center-line Velocity for the Fluent Model and Experimental Results

Fig. 1 shows the velocity at the center-line for both the fluent model and the experimental model. The Fluent model roughly follows the same trend as the experimental model with the results converging with increasing x/L (farther from the flame inlet).

Table 1 Percent Difference in Axial Velocity Between Experimental and Fluent Data

x/L	0	1/16	1/8	1/4	3/8	1/2	5/8	3/4	1
Percent Difference	NaN	30.9319	35.7702	31.9602	31.4705	27.1014	NaN	17.2525	9.9835

Table 1 shows the percent differences of the axial velocity at differing locations on the horizontal axis. It can easily be seen that the location of 1/8 has the largest percent difference then they steadily get smaller which corresponds with Fig. 1. It is important to note the NaN corresponds to where there is not experimental data thus a percent difference cannot be calculated.

2 Fluent Model and Experimental Data Comparison - Q3

The $k - \epsilon$ Standard turbulent model is used to compute an array of differing outputs at different axial and lengthwise locations. All the default constants are used in this case ($C_{\epsilon 1} = 1.44$).

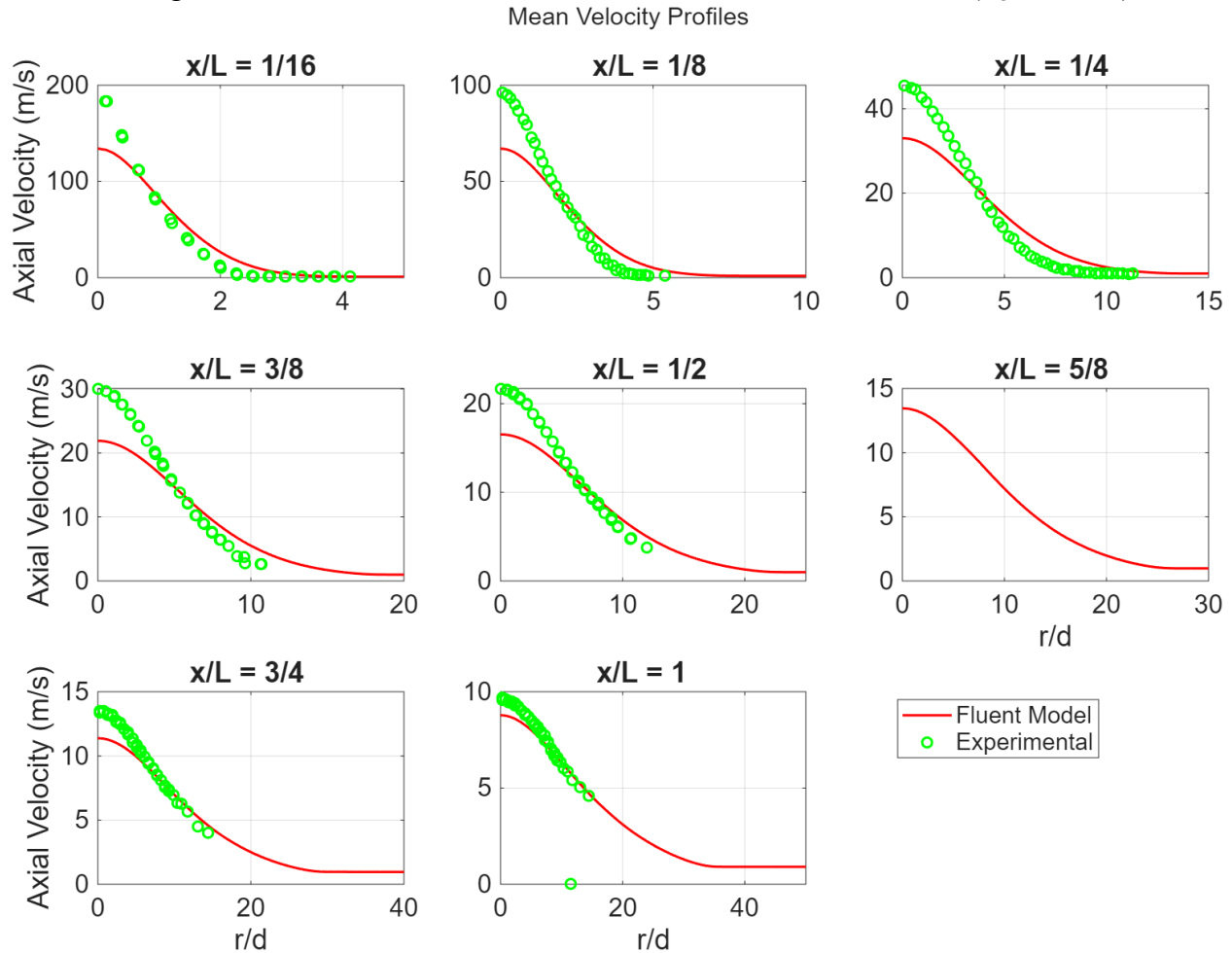


Figure 2 Axial Velocity Profile

Fig. 2 shows the axial velocity profile of the fluent model and experimental results at differing horizontal locations (x/L). It can be noted that the experimental results follow the same trend as

the model; however, the centerline location ($r/d=0$) has significantly larger experimental results compared to the model than locations farther from the flame in the radial direction.

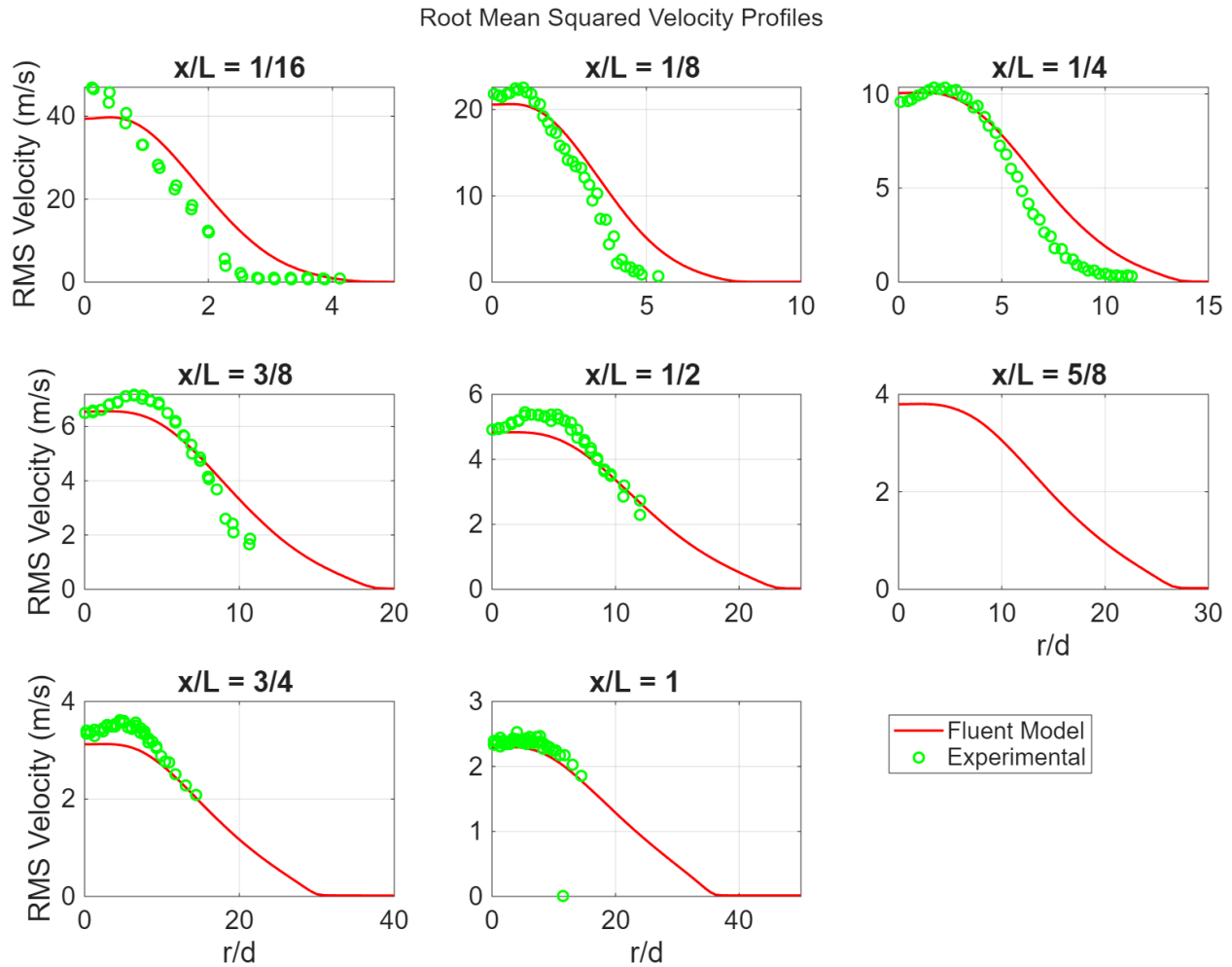


Figure 3 Root Mean Squared Velocity Profile

Fig. 3 shows the RMS velocity profile. It shows the same trends as Fig. 2 does; however, the experimental results match better with the fluent model. Near the center-line, the experimental results are slightly higher than they are supposed to be; however, they match the model well near the middle of the radial direction. As there are increases in the radial direction, the fluent model tends to overestimate the experimental data instead of underestimating it.

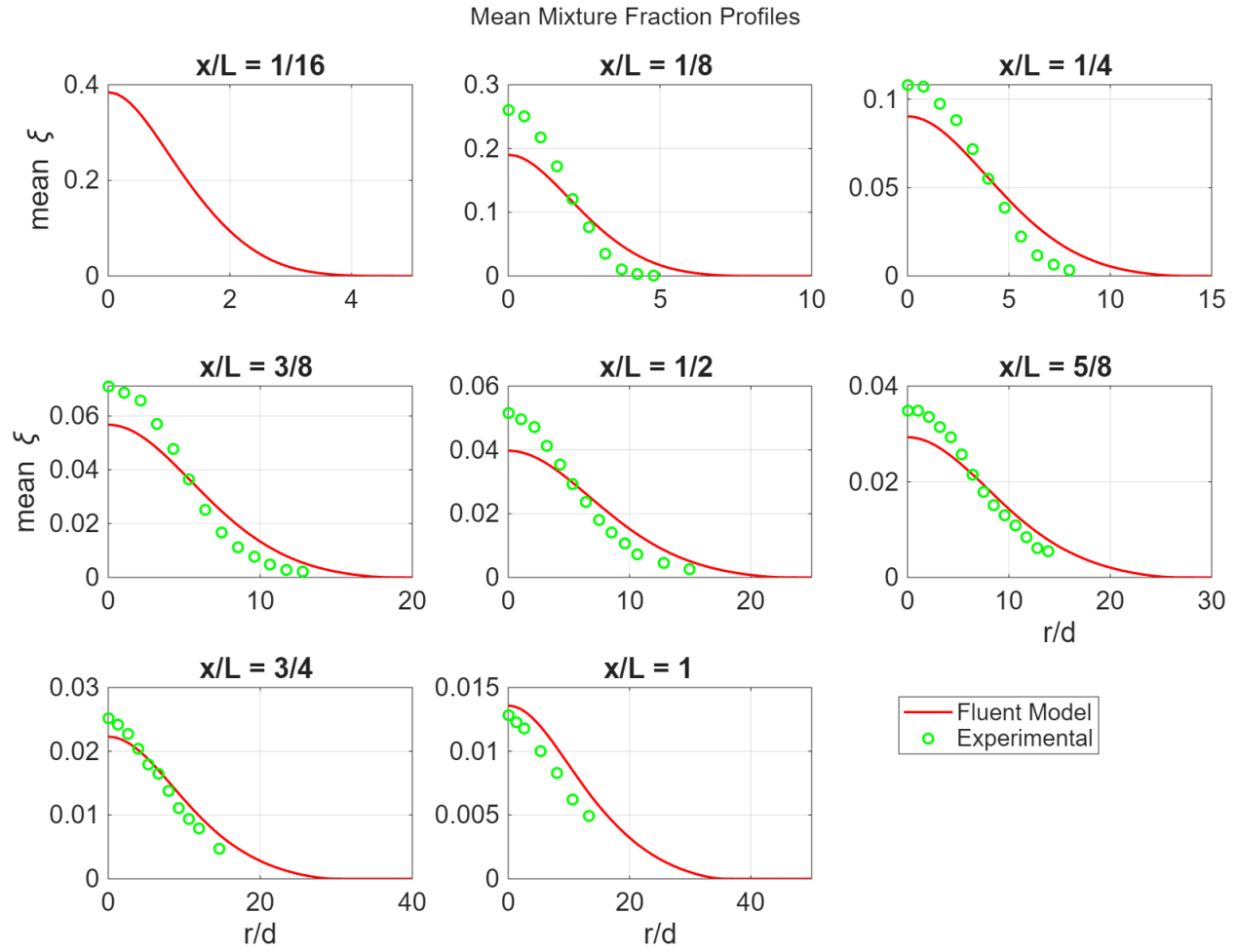


Figure 4 Mean Mixture Fraction

Fig. 4 shows the mean mixture fraction at varying horizontal distances comparing the fluent model and the experimental results. The model and experimental results match up not as well as closer to the inlet, but as the flame travels down the tube, the experimental results and the model converge more. The model tends to underestimate the data; however, as the radial direction increases, the model tends to overestimate the data instead.

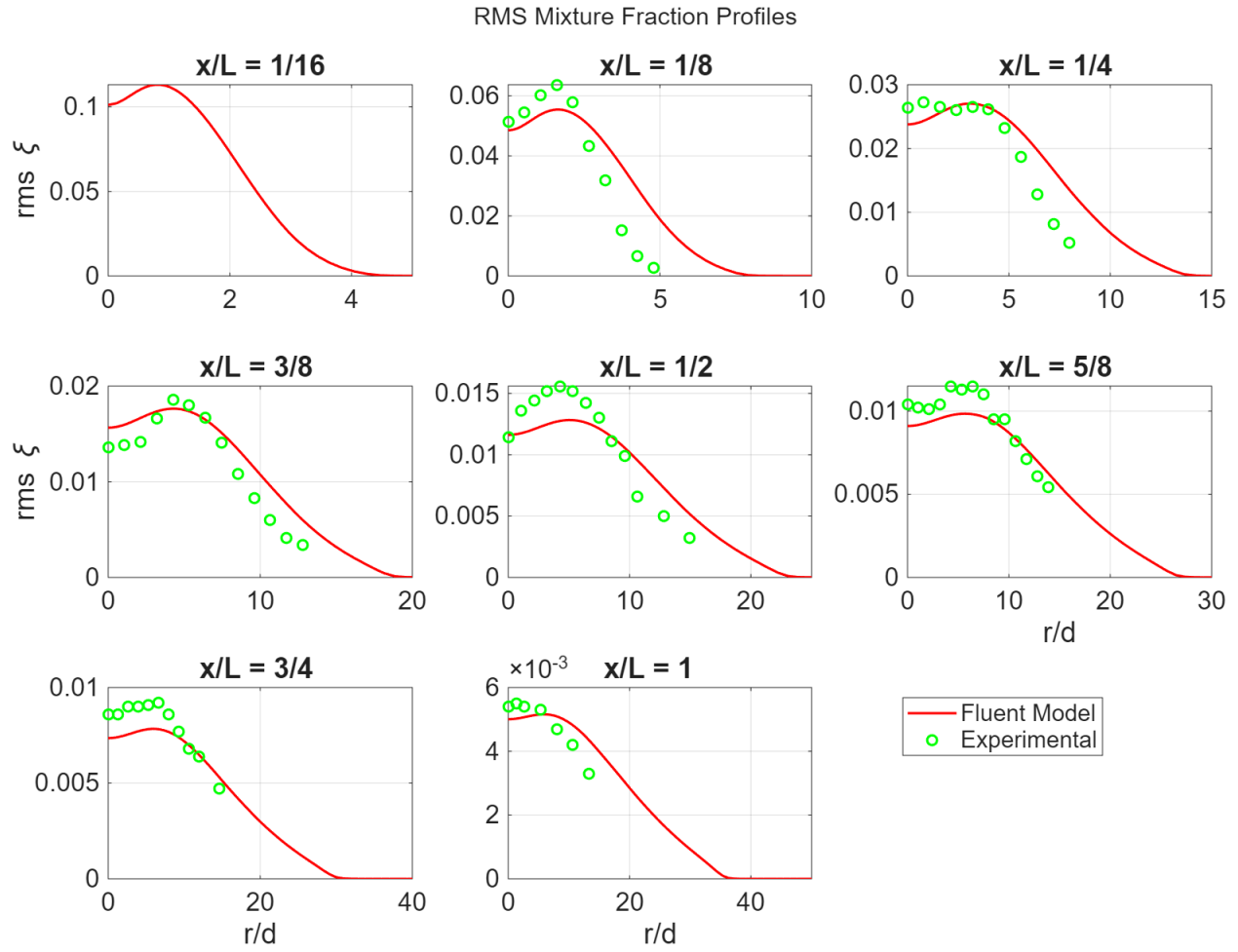


Figure 5 Root Mean Squared Mixture Fraction

Fig. 5 shows the root mean squared of the mixture fraction at varying horizontal locations comparing both the fluent model and the experimental values. The fluent model tends to underestimate the results, especially where the RMS mixture fraction is at a peak. However, as there are increases in the radial direction, the fluent model tends to overestimate the data.

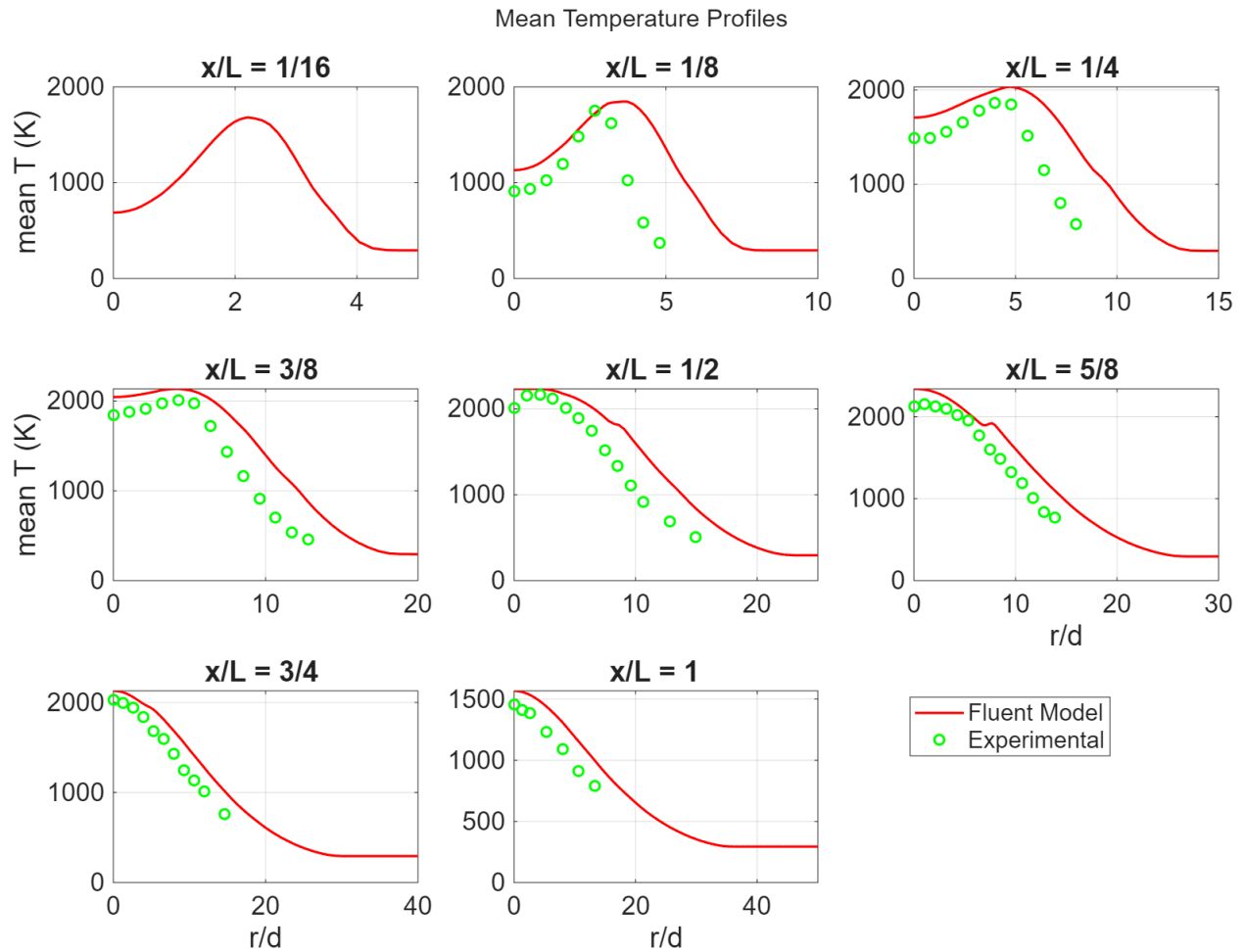


Figure 6 Mean Temperature

Fig. 6 shows the mean temperature at varying horizontal locations comparing the fluent model and experimental results. The model tends to consistently overestimate the experimental data. The biggest gaps in the model and the experimental results occurs where there are large temperature changes such as near the flame inlet on the edge of the flame. These are shown in x/L location of $1/8$ and $1/4$ where the temperature drastically decreases.

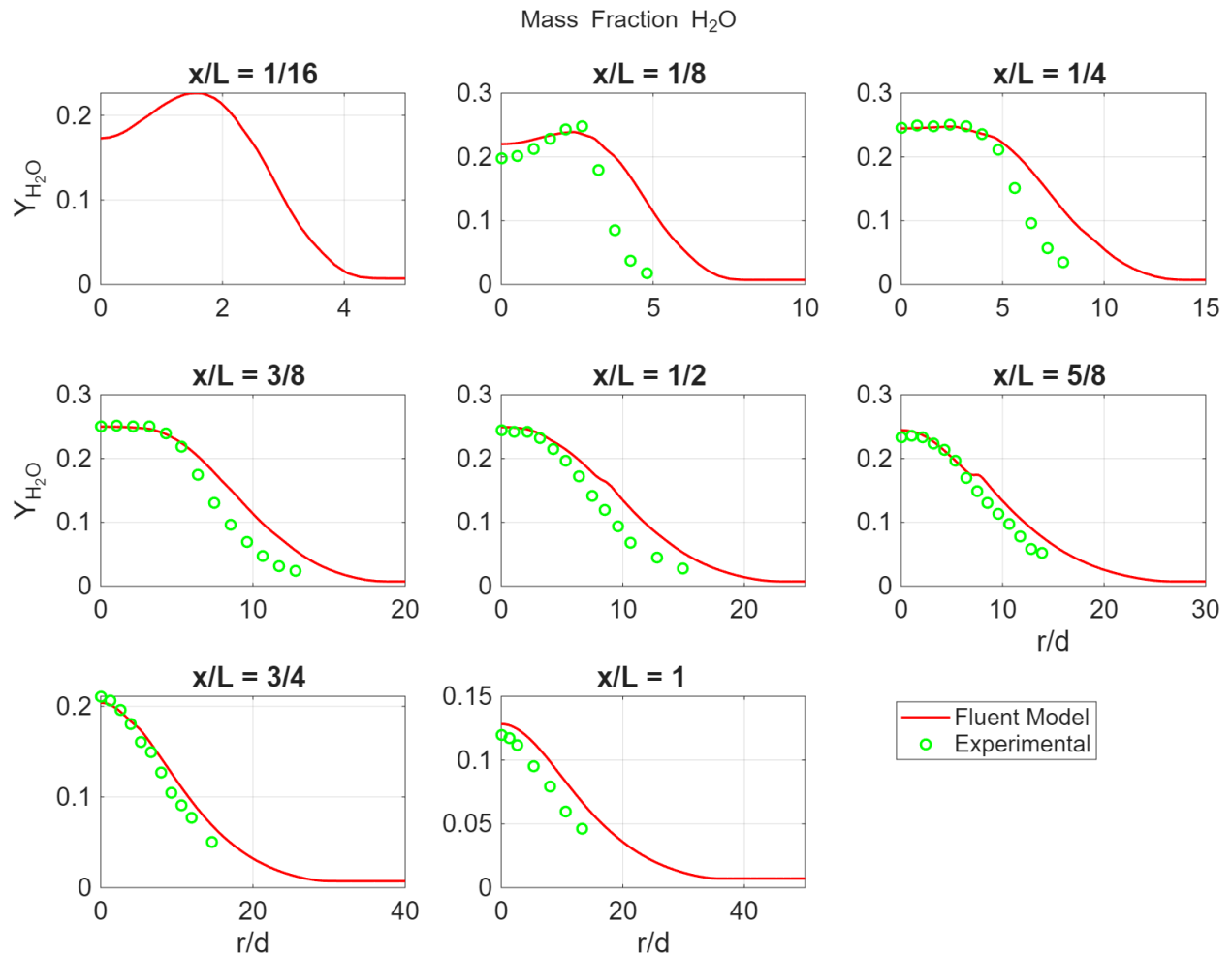


Fig. 7 shows the mass fraction of H_2O at differing horizontal locations. The fluent model and experimental data match up well except for near the inlet where there is more deviation near the edge of the flame. However, this is to be expected as this is where there is the most drastic change. The model tends to consistently overestimate the mass fraction of the experimental data.

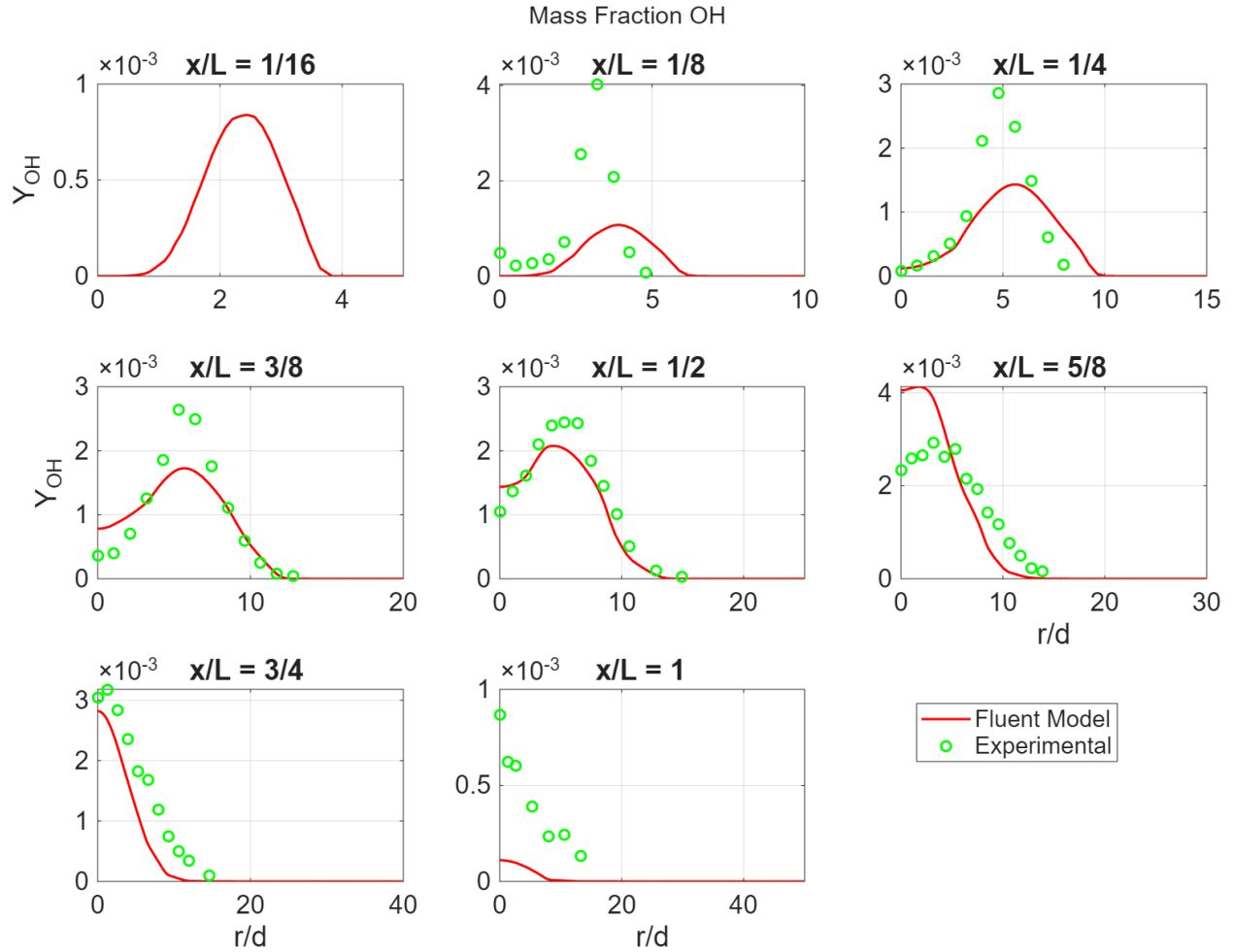


Figure 8 Mass Fraction of OH

Fig. 8 shows the mass fraction of OH as the horizontal distance varies. Similar to the mass fraction of H_2O , there are more differences near the flame inlet. However, there are significant differences in the model and the experimental data. The model almost always underestimates the experimental data, especially near the edges of the flame near the inlet. This makes sense as OH is a radical which can make it harder to predict as its presence is more fleeting.

3 Dependence of the Calculations on the Turbulence Model Constants – Q4

The $k - \epsilon$ Standard turbulent model was used to compute an array of differing outputs at different axial and lengthwise locations. In this case, the $C_{\epsilon 1}$ constant is changed from the default 1.44 to 1.6, and the changes in the model results are compared with the experimental results.

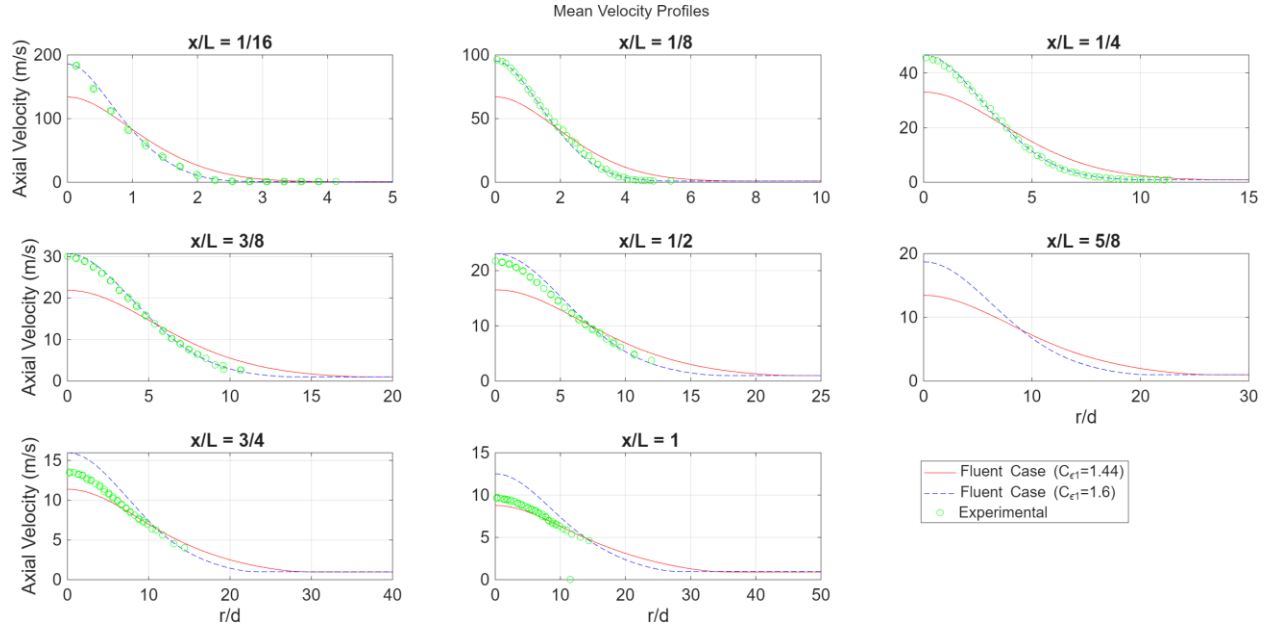


Figure 9 Mean Velocity

Fig. 9 compares the axial velocity of the two fluent cases to the experimental data. The experimental data is between both the $C_{\epsilon 1}$ models, and as the horizontal distance increases, the better the $C_{\epsilon 1} = 1.44$ case models the data and the $C_{\epsilon 1} = 1.6$ is a better model near the inlet.

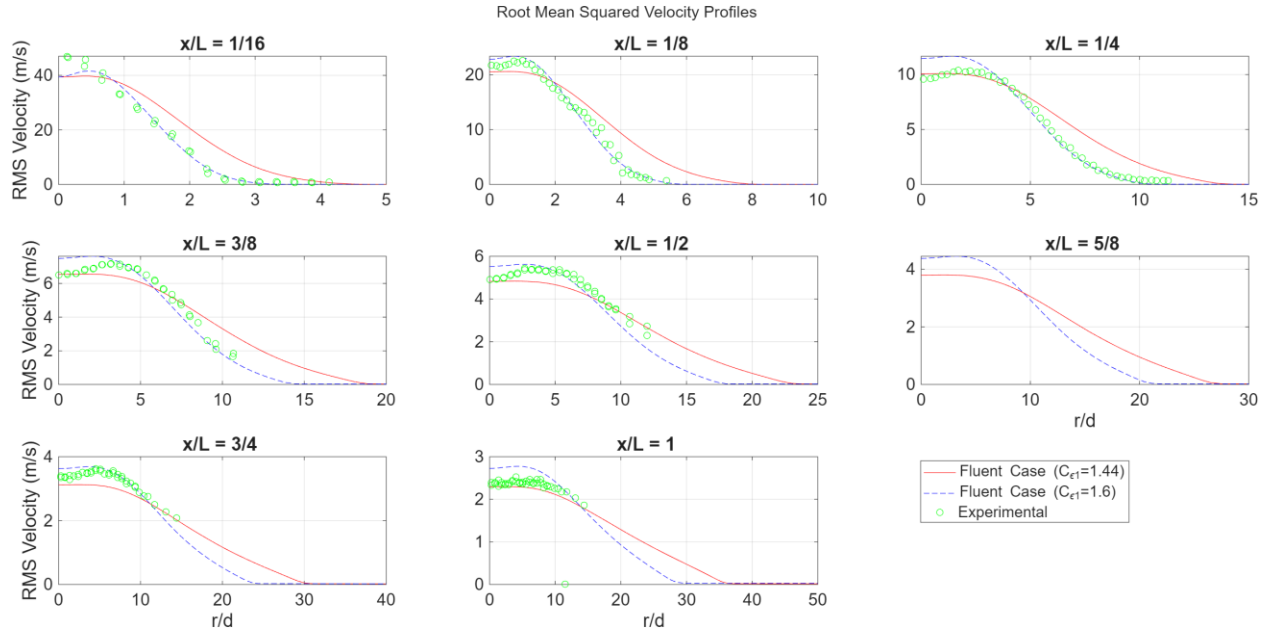


Figure 10 Root Mean Square Velocity

Fig. 10 shows the root mean squared velocity at differing horizontal locations. The experimental data is between both the $C_{\epsilon 1}$ models for all of the locations except for right where the fuel inlet is. The $C_{\epsilon 1} = 1.44$ model tends to be a better fit near the centerline, and the $C_{\epsilon 1} = 1.6$ model tends to be a better fit as the radial position increases.

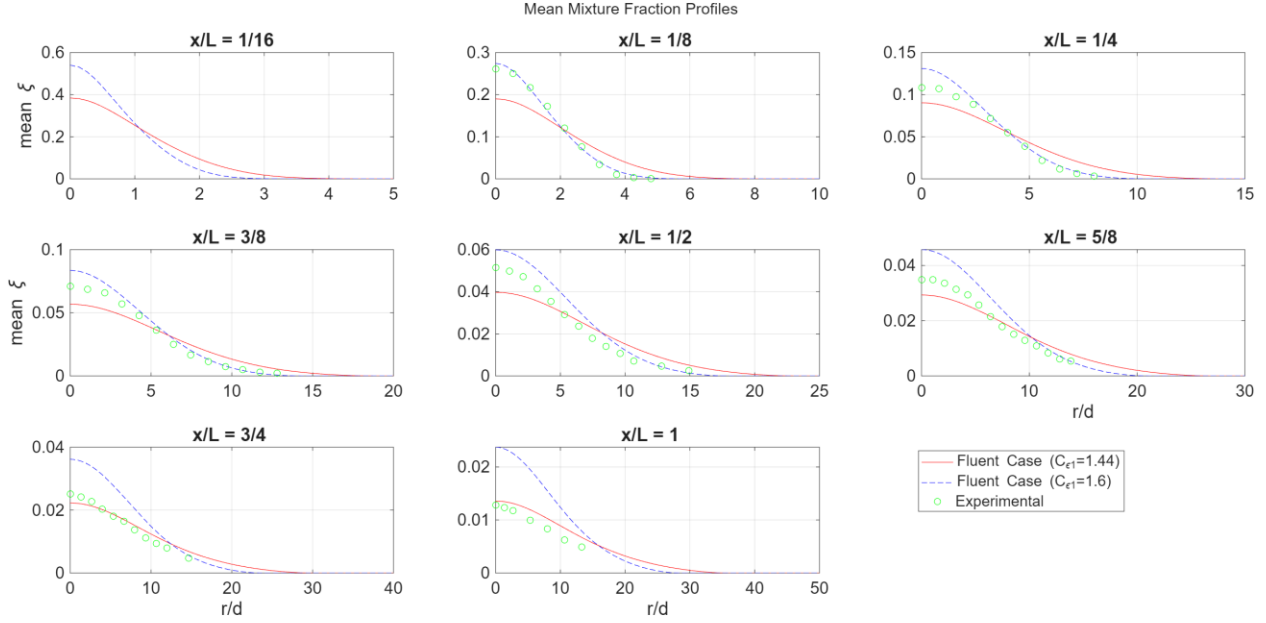


Figure 11 Mean Mixture Fraction

Fig. 11 shows the mean mixture fraction at differing horizontal locations. The experimental data is between the C_{ϵ_1} models for all the locations except for at the flame outlet where both the models overestimate the results. The $C_{\epsilon_1} = 1.6$ model tends to be a better fit near the fuel inlet, and the $C_{\epsilon_1} = 1.44$ model tends to be a better fit as the horizontal position increases.

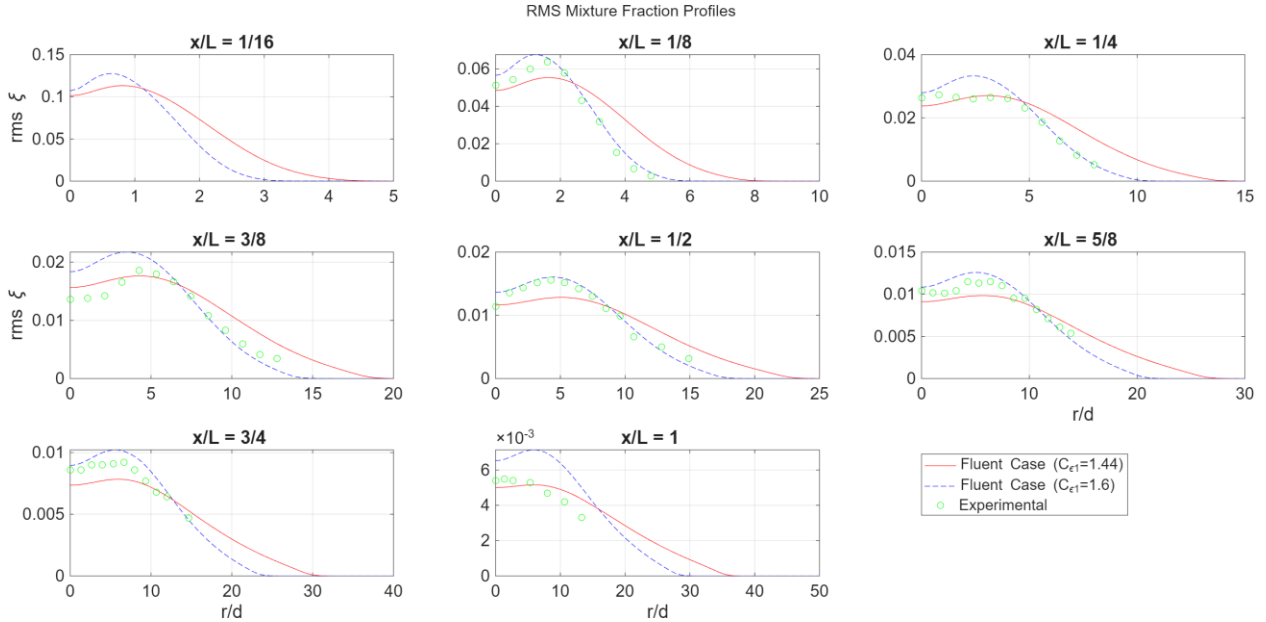


Figure 12 Root Mean Square Mixture Fraction

Fig. 12 shows the RMS mixture fraction at differing horizontal locations. The experimental data is between the C_{ϵ_1} models for almost all the locations. The $C_{\epsilon_1} = 1.6$ model tends to be a better fit near the fuel inlet, and the $C_{\epsilon_1} = 1.44$ model tends to be a better fit as the horizontal position increases. However, the exception for this rule is at the $x/L=3/8$ location where the $C_{\epsilon_1} = 1.44$ is a much better fit even though this model overestimates the results.

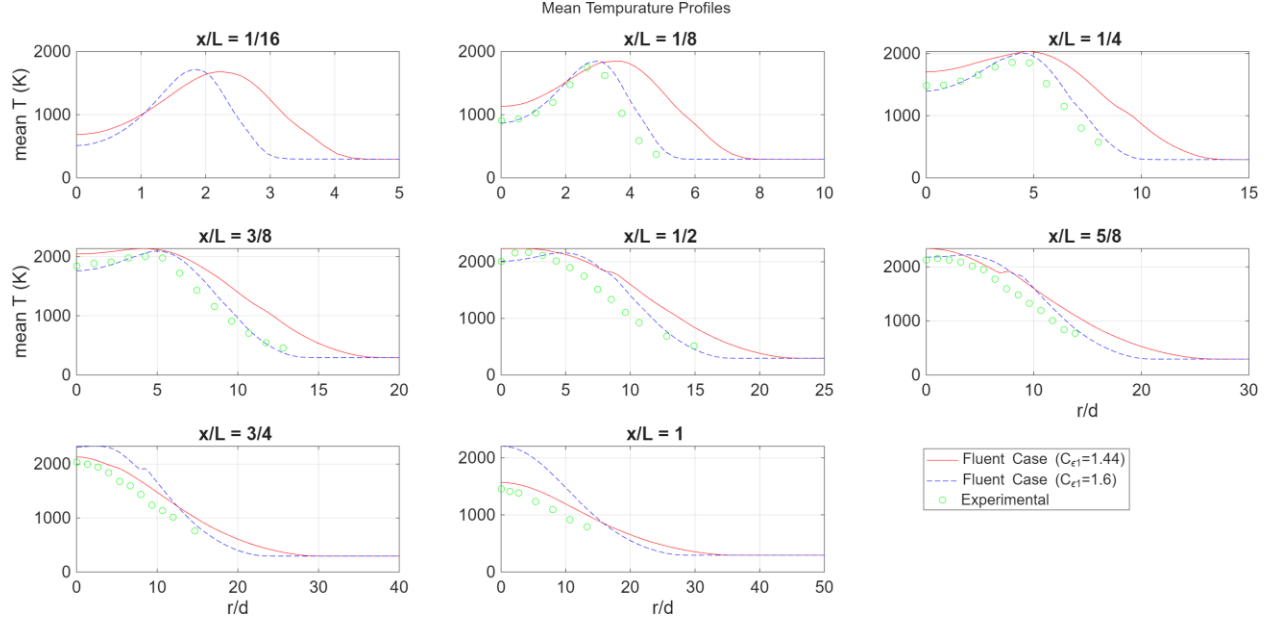


Figure 13 Mean Temperature

Fig. 13 shows the mean temperature at differing horizontal locations. The experimental data is below the $C_{\epsilon 1}$ models for all the locations. The better fitting model is whatever one predicts the lowest temperature as the models consistently overestimate the experimental results.

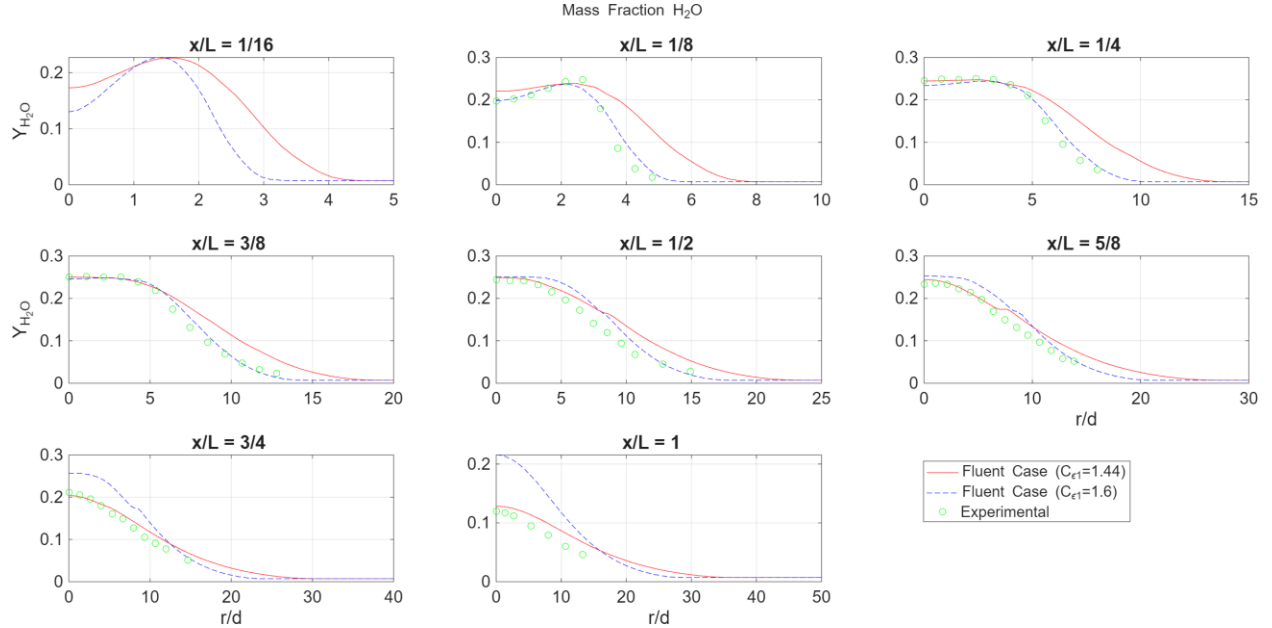


Figure 14 Mass Fraction of H_2O

Fig. 14 shows the mass fraction of H_2O at differing horizontal locations. The experimental data is below the $C_{\epsilon 1}$ models for almost all the locations. The better fitting model is whatever one predicts the lowest mass fraction as the models consistently overestimate the experimental results. This tends to be the $C_{\epsilon 1} = 1.6$ case near the inlet and the $C_{\epsilon 1} = 1.44$ case as the horizontal location increases.

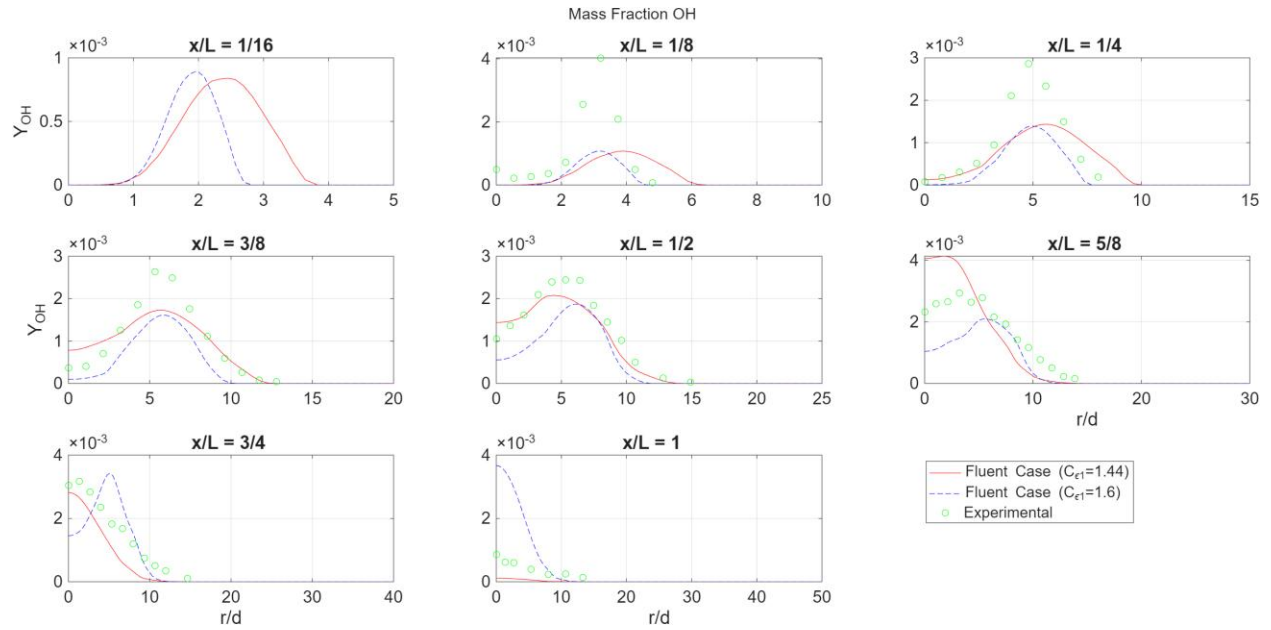


Figure 15 Mass Fraction of OH

Fig. 15 shows the mass fraction of OH at differing horizontal locations. Near the fuel inlet, the models greatly underestimate the experimental results while this trend decreases as horizontal location increases. In the middle of the tube (horizontally speaking), the models tend to predict the results slightly better. As the horizontal location increases, the $C_{\epsilon 1} = 1.44$ model does a better job of predicting the shape while near the inlet the $C_{\epsilon 1} = 1.6$ case does a better job.

Table 2 Maximum Percent Difference for Each Data Output at Differing x/L for $C_{\epsilon 1} = 1.44$ and Experimental

x/L	1/16	1/8	1/4	3/8	1/2	5/8	3/4
Axial Velocity	36.1922	23.5633	57.6312	35.7702	63.14109	NaN	29.5048
RMS Velocity	19.0097	11.8777	30.7140	17.6297	31.7387	NaN	19.5505
Mean ξ	NaN	52.8911	36.4858	85.2634	31.4300	66.9619	42.1047
RMS ξ	NaN	17.8647	22.5245	24.6163	15.6247	18.9500	19.9906
Mean Temperature	NaN	34.8177	23.1592	56.8312	21.7738	37.8168	30.5325
Mass Fraction H_2O	NaN	49.2743	34.3171	80.2010	10.6009	50.7401	40.7622
Mass Fraction OH	NaN	191.0481	176.6545	199.3098	198.7189	198.8341	199.8525

Table 2 shows maximum percent differences among differing radial values with constant distances from inlet for the $C_{\epsilon 1} = 1.44$ case and the experimental values. The percent differences are much larger for the mass fraction of the OH especially near the outlet. This model tends to do a better job predicting the values near the outlet due to the lower percent differences; however, that is not always the case. There are large percent differences with the values ranging from 11% to 200% showing that this model does not accurately describe the experimental results well.

Table 3 Maximum Percent Difference for Each Data Output at Differing x/L for $C_{\epsilon 1} = 1.6$ and Experimental

x/L	1/16	1/8	1/4	3/8	1/2	5/8	3/4
Axial Velocity	67.6331	31.1132	57.6312	25.3352	63.1411	NaN	29.5048
RMS Velocity	35.8820	22.2408	30.7140	17.7154	31.7387	NaN	19.5505
Mean ξ	NaN	100.1210	30.2361	85.2634	59.8043	66.9619	42.1047
RMS ξ	NaN	23.0386	28.9224	24.6163	29.7664	18.9500	19.9906
Mean Temperature	NaN	66.5574	27.0787	56.8312	40.9389	37.8168	30.5325
Mass Fraction H_2O	NaN	94.0945	26.6212	80.2010	56.8935	50.7401	40.7622
Mass Fraction OH	NaN	199.9322	198.1020	199.3098	199.9800	198.8341	199.8525

Table 3 shows maximum percent differences among differing radial values with constant distances from inlet for the $C_{\epsilon 1} = 1.6$ case and the experimental values. The percent differences are much larger for the mass fraction of the OH as they are consistently around 200%. This model tends to do a better job predicting the values near the outlet due to the lower percent differences; however, that is not always the case. The percent differences range from 17% to 200% showing that this model is not that accurate.

Comparing both Table 2 and Table 3, it can be concluded that the models do a better job near the outlet which makes sense as this is where there is less discontinuity from the flame. These tables show the maximum percent difference. This tends to be near the edge of the flame which accounts for the large percent differences. The $C_{\epsilon 1}$ value does matter as the models are significantly different with the differing value. Overall, the models hold the same shape, so they predict similar trends even though the numbers can be quite different.

4 Performance of Different Turbulence Models in Fluent – Q5

The differing turbulence models in fluent all calculate the output values slightly differently. This section focuses on the difference between the differing $k - \epsilon$ turbulence models. The default $C_{\epsilon 1}$ value for the standard model is 1.44, and the default $C_{\epsilon 1}$ value for the RNG model is 1.42. Linear interpolation was done to find the target $C_{\epsilon 1}$ values for each model to match the

calculated mean velocity at the center-line at an x/L value of $1/2$ to the experimental data at this location. Eq. (1) and Eq. (2) show the equations to accomplish this.

$$\frac{du}{dC_{\epsilon 1}} = \frac{U_2 - U_1}{C_{\epsilon 1,2} - C_{\epsilon 1,1}} \quad (1)$$

$$C_{\epsilon 1, new} = C_{\epsilon 1,1} + \frac{U_{ex} - U_1}{\frac{du}{dC_{\epsilon 1}}} \quad (2)$$

The U values are the mean velocities with the first state being the values that correspond with the $C_{\epsilon 1} = 1.44/1.42$ model and the second state being the values that correspond with the $C_{\epsilon 1} = 1.6$ model. The U_{ex} value is the experimental velocity at the specified location.

Table 4 Mean (Axial) Velocity on the Centerline ($x/L = 1/2$) (m/s)

	Experimental Velocity	Calculated Velocity with Default Values	Calculated Velocity with $C_{\epsilon 1} = 1.6$	Target $C_{\epsilon 1}$ Value Calculated	Calculated Velocity at Target
Standard	21.7126	16.5304	23.0843	1.5665	21.3054
RNG	21.7126	11.5786	14.4475	1.821	21.4360

Table 4 shows the values for this process. A fluent model was run for both cases at a $C_{\epsilon 1} = 1.44/1.42$ case and a $C_{\epsilon 1} = 1.6$ case and the axial velocities were recorded. Eq. (1) and Eq. (2) were then used to solve the new $C_{\epsilon 1}$ values, then another case was run. The percent difference for the experimental velocity and calculated velocity at the target $C_{\epsilon 1}$ is 1.893% and 1.2821% for the standard model and RNG model respectively.

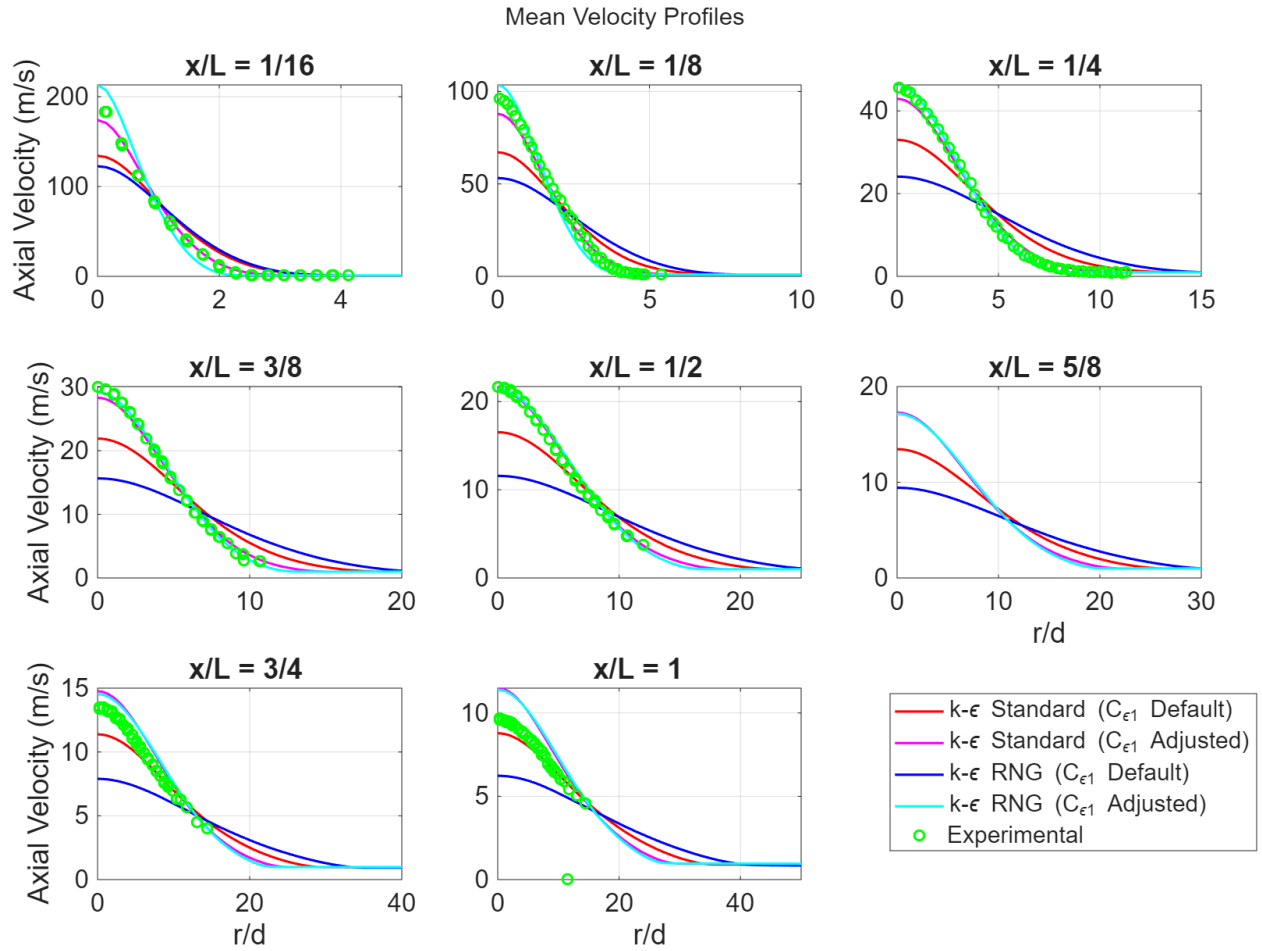


Fig. 16 shows the mean velocity at different horizontal locations comparing the different models and $C_{\epsilon 1}$ values. The RNG models follow each other more closely than the standard model. The RNG model also predicts a lower axial velocity compared to the standard model and the $C_{\epsilon 1}$ adjusted models. The experimental data follows the adjusted models well as this is what they are based on.

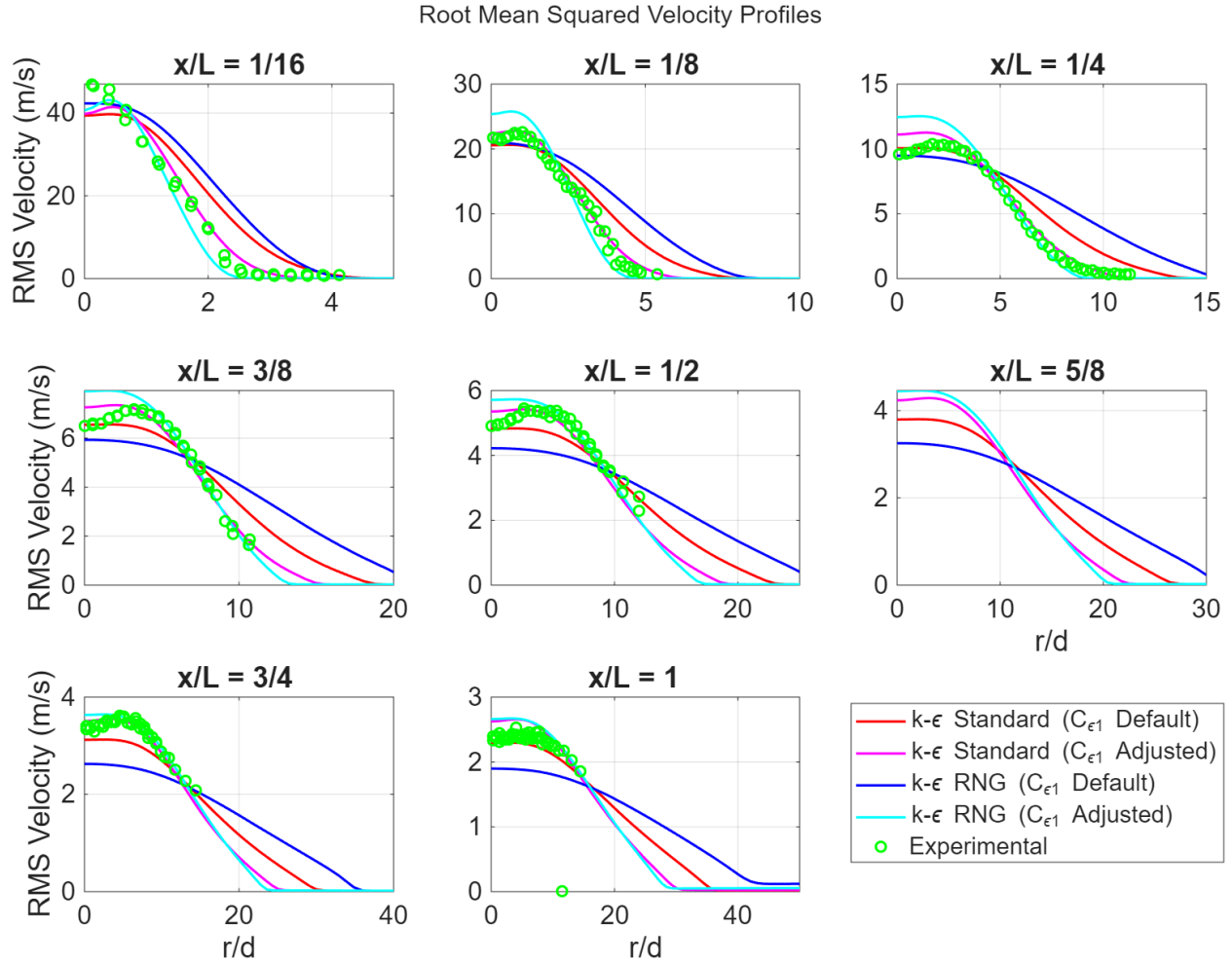


Figure 17 Root Mean Squared Velocity

Fig. 17 shows the root mean square velocity for differing horizontal locations comparing the different models and experimental data. The standard $C_{\epsilon 1}$ adjusted model tracks the experimental data the best and is thus the better model. The RNG $C_{\epsilon 1}$ default model tracks the experiential data the worst as it predicts the flattest results, and the model consistently predicts lower values near the centerline and higher values near the radial edge. The $C_{\epsilon 1}$ adjusted models are very consistent as they are calibrated to each other.

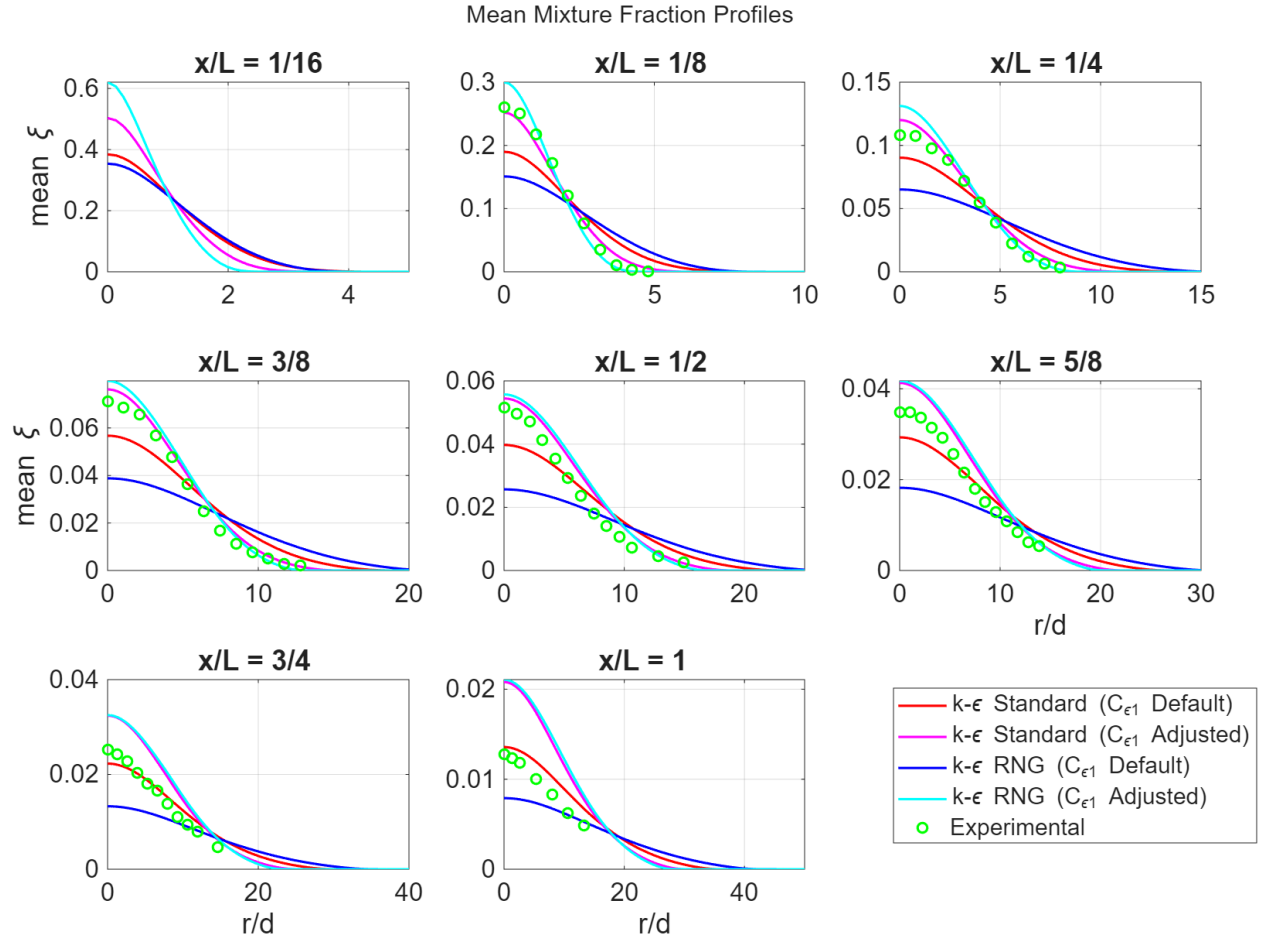


Figure 18 Mean Mixture Fraction

Fig. 18 shows the mean mixture fraction for differing horizontal locations comparing the different models and experimental data. Near the inlet, the standard $C_{\epsilon 1}$ adjusted model provides the best results while the standard $C_{\epsilon 1}$ default model provides the best results near the outlet when compared to the experimental values. The $C_{\epsilon 1}$ adjusted models track each other well which predict higher results compared to experimental models. The RNG $C_{\epsilon 1}$ default model predicts the flattest results, and the model consistently predicts lower values near the centerline and higher values near the radial edge.

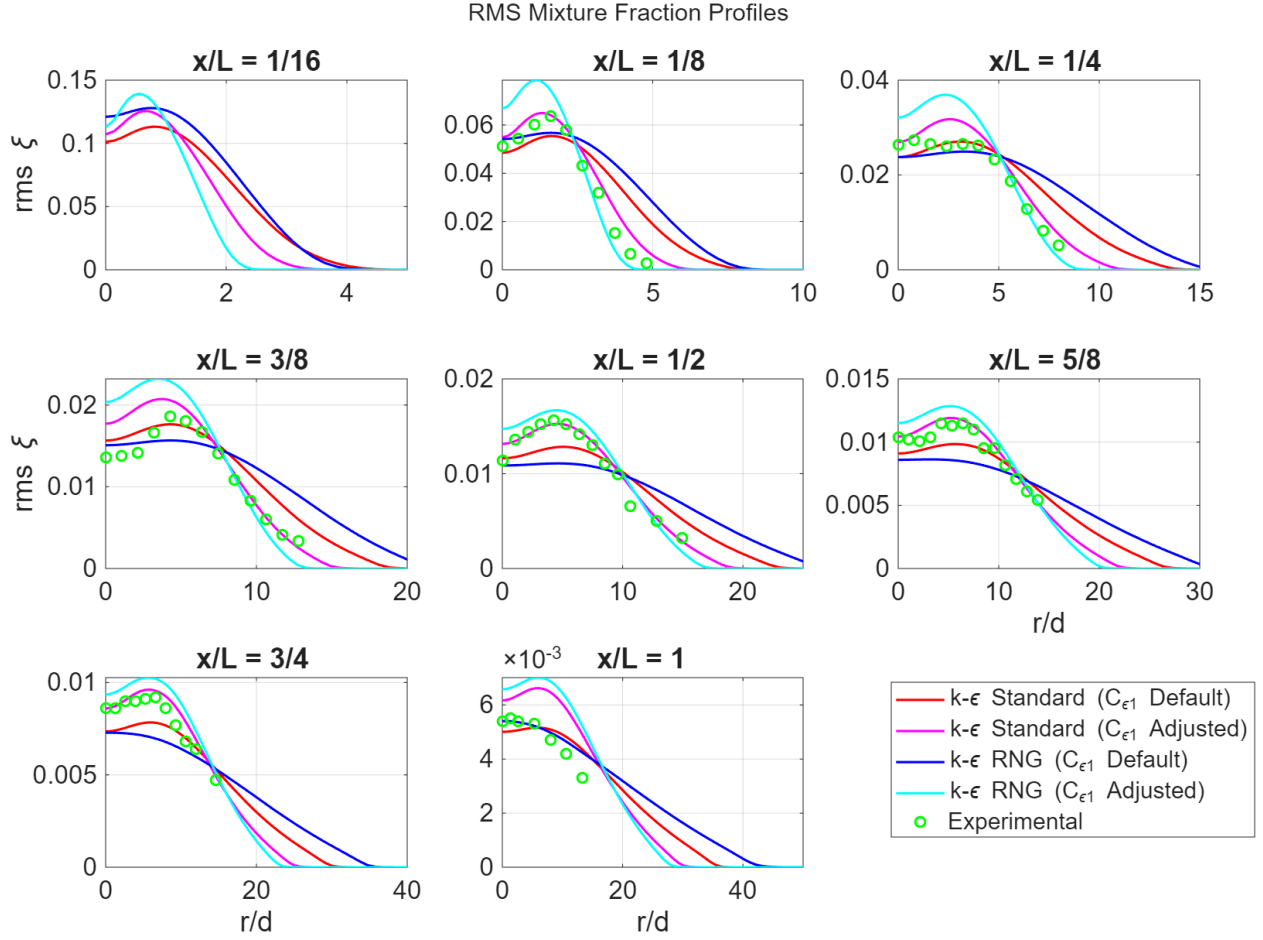


Figure 19 RMS Mixture Fraction

Fig. 19 shows the root mean squared mixture fraction for differing horizontal locations comparing the different models and experimental data. The standard $C_{\epsilon 1}$ adjusted model does the best at modeling the experimental data compared to the other models except for near the outlet which follows the standard $C_{\epsilon 1}$ default model better. The RNG $C_{\epsilon 1}$ default model is the outlier among the models as it predicts flatter results and small mixture fractions compared to the other models. Other than that. The models are similar with the RNG $C_{\epsilon 1}$ adjusted model predicting the largest values.

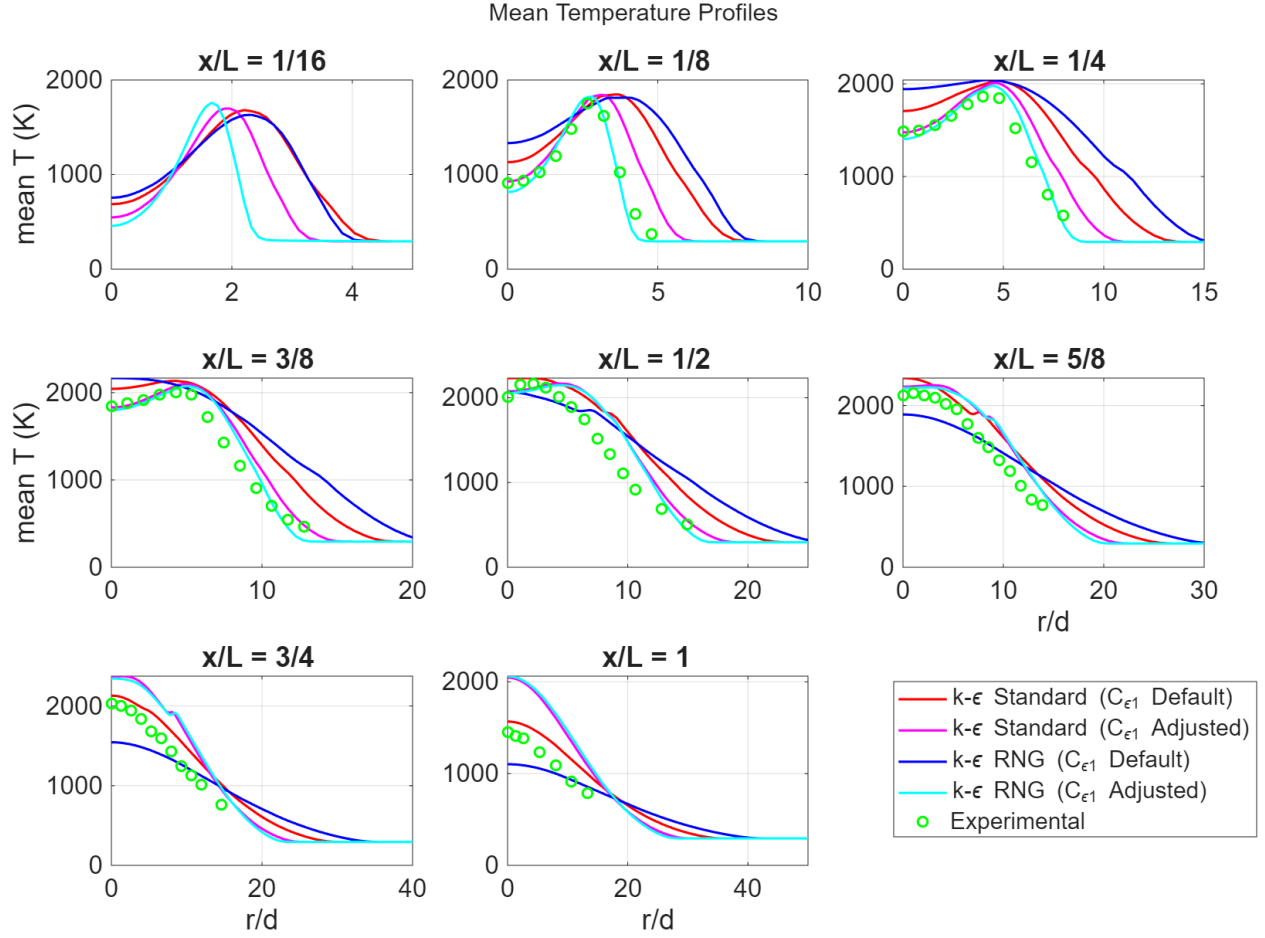


Figure 20 Mean Temperature

Fig. 20 shows the mean temperature for differing horizontal locations comparing the different models and experimental data. The models predict the experimental model well with the RNG $C_{\epsilon 1}$ adjusted model doing the best near the inlet and the standard $C_{\epsilon 1}$ default model doing better near the outlet when compared to the experimental values. The RNG $C_{\epsilon 1}$ default model does not align as closely to the other models as it consistently predicts lower values near the centerline and higher values near the radial edge.

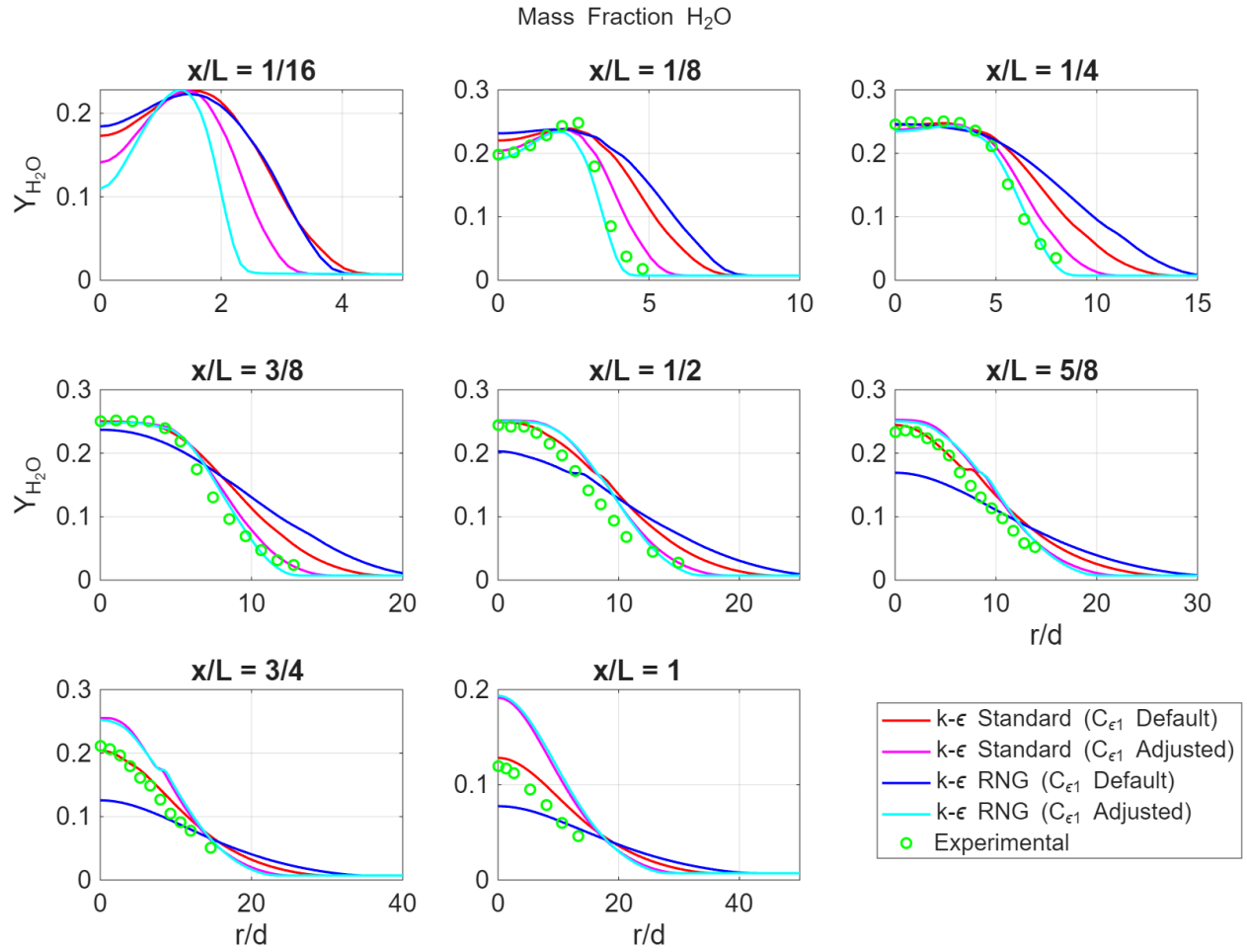


Figure 21 H₂O Mass Fraction

Fig. 21 shows the mass fraction of H₂O for differing horizontal locations comparing the different models and experimental data. The models do a decent job of predicting experimental data with the RNG $C_{\epsilon 1}$ adjusted model doing better at the inlet and the standard $C_{\epsilon 1}$ default model doing better at the outlet. The models tend to match each other well except for the RNG $C_{\epsilon 1}$ default model which consistently predicts lower values near the centerline and higher values near the radial edge. The RNG $C_{\epsilon 1}$ default and adjusted models vary drastically in their results as well as the standard $C_{\epsilon 1}$ default and adjusted models coincide better.

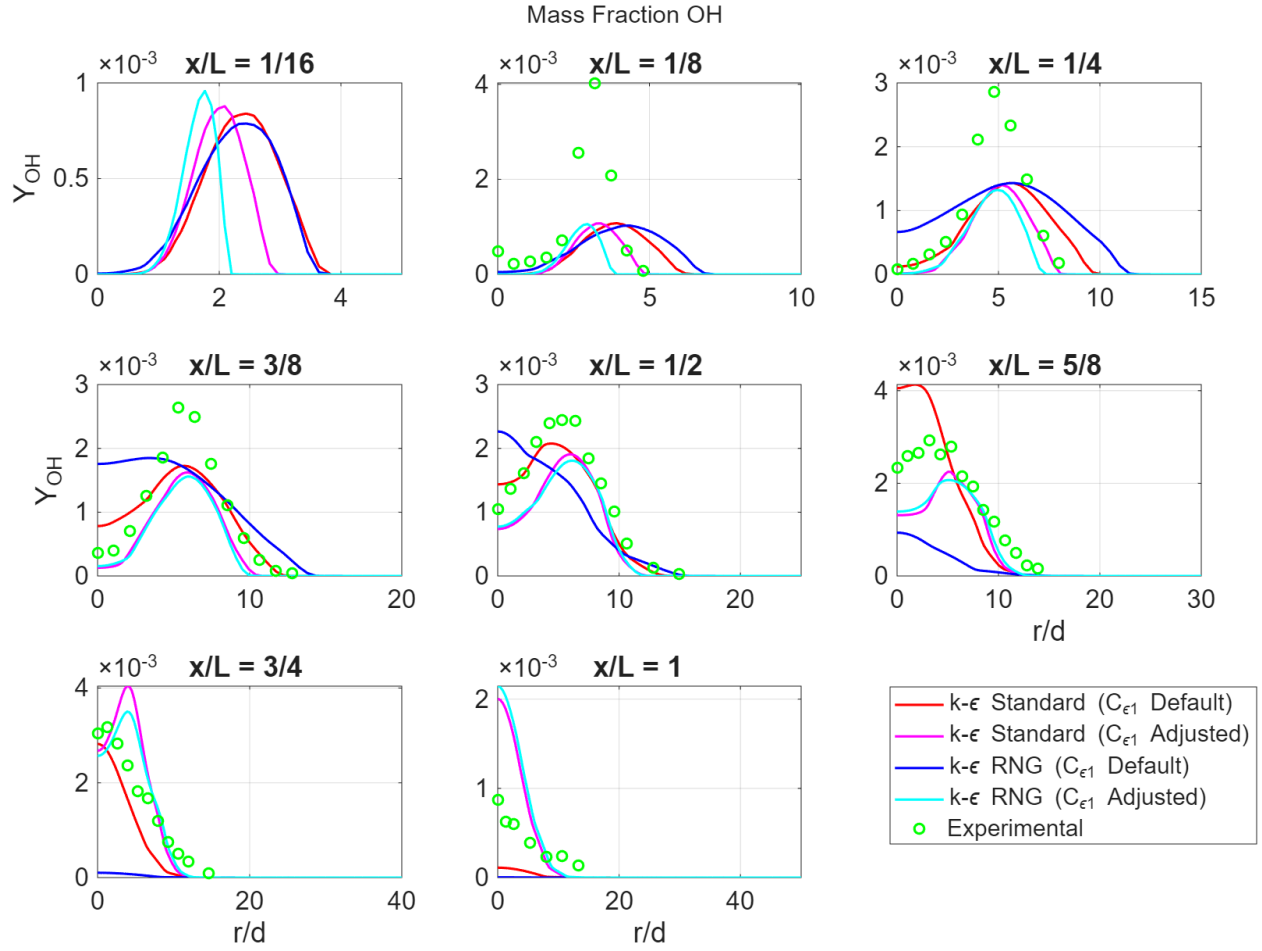


Figure 22 OH Mass Fraction

Fig. 22 shows the mass fraction of OH for differing horizontal locations comparing the different models and experimental data. The models do not do a good job of predicting the experimental results near the inlet, but it gets better near the outlet. The $C_{\epsilon 1}$ adjusted results do a better job of predicting the trend of the data than the $C_{\epsilon 1}$ default models. There are major differences in the RNG $C_{\epsilon 1}$ default model compared to all the other models as it has different trends and values consistently.

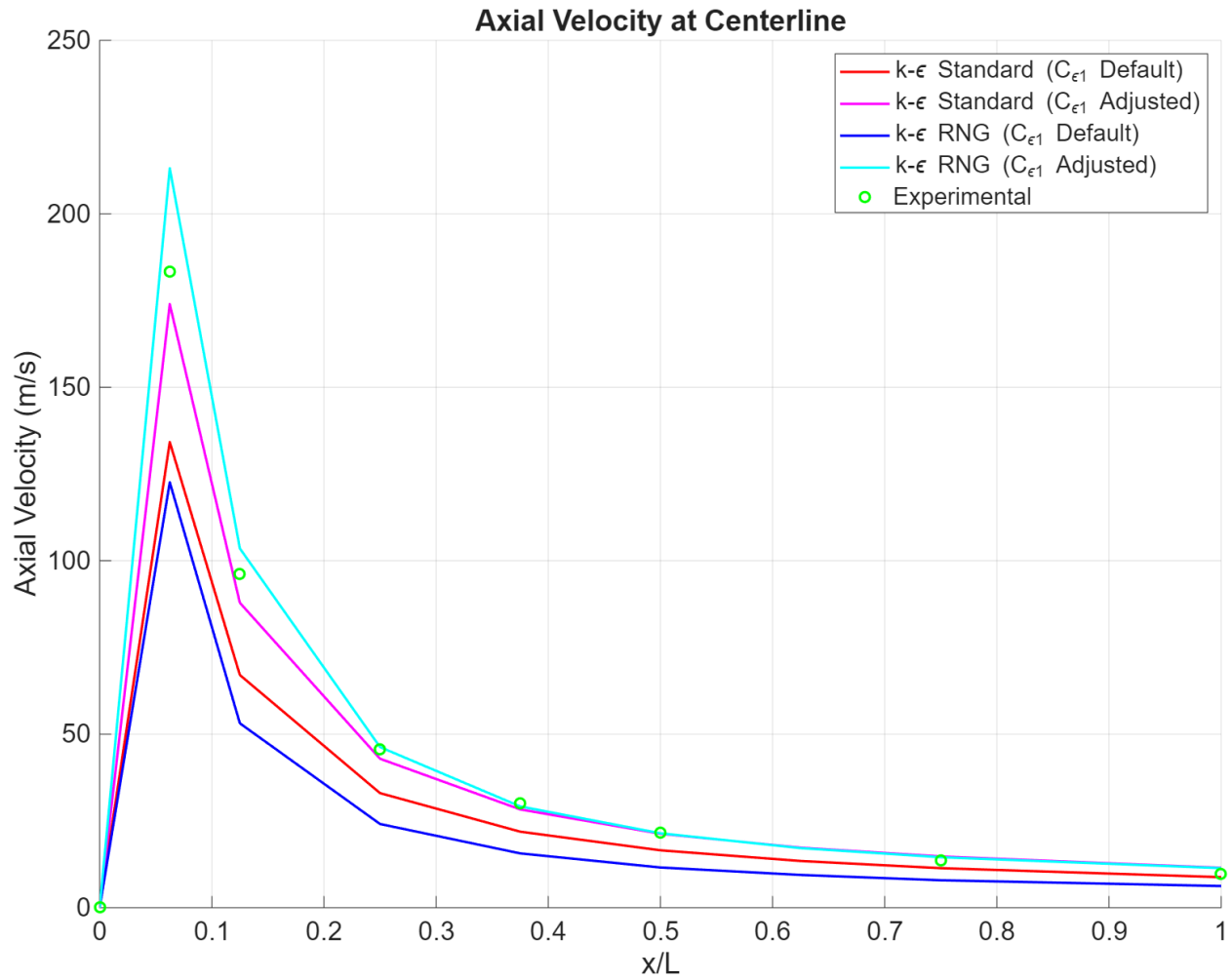


Figure 23 Axial Velocity at Centerline

Fig. 23 shows the axial velocity at the centerline for each of the models and experimental data. The differing models differ greatly near the inlet, and they converge slightly with the $C_{\epsilon 1}$ adjusted models converging the most near the outlet. The standard $C_{\epsilon 1}$ adjusted model concedes the best with the experimental data. The $C_{\epsilon 1}$ default models do not do as good of a job predicting the axial velocity as the models underestimate the data.

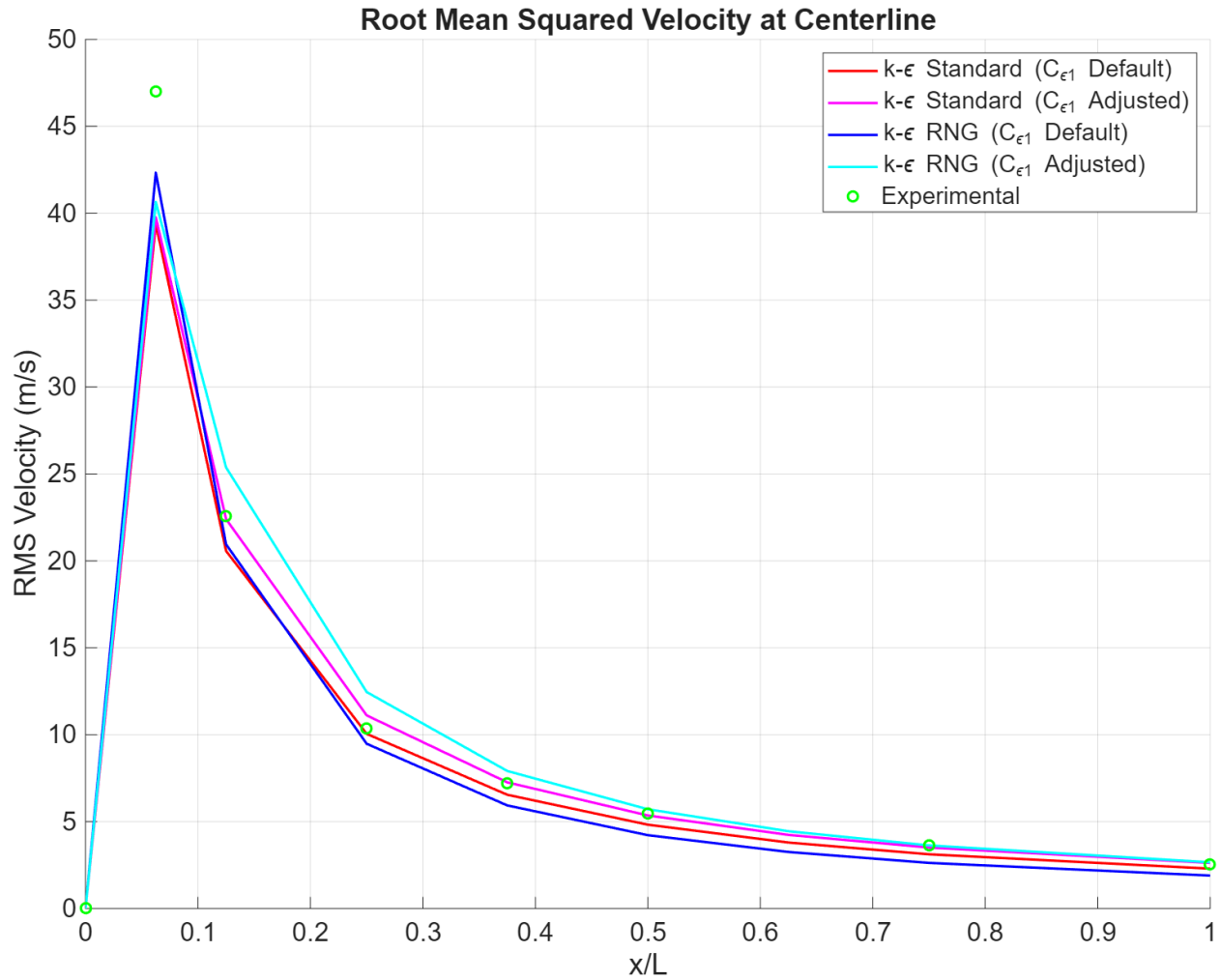


Figure 24 Root Mean Square Velocity at Centerline

Fig. 24 shows the RMS velocity at the centerline for each of the models and experimental data. The differing modes do match well at the inlet and the outlet with the RNG $C_{\epsilon 1}$ default value predicting the highest value and the standard $C_{\epsilon 1}$ default predicting the lowest value; however, all these values are very close. The standard $C_{\epsilon 1}$ adjusted model matches the experimental data the best consistently.

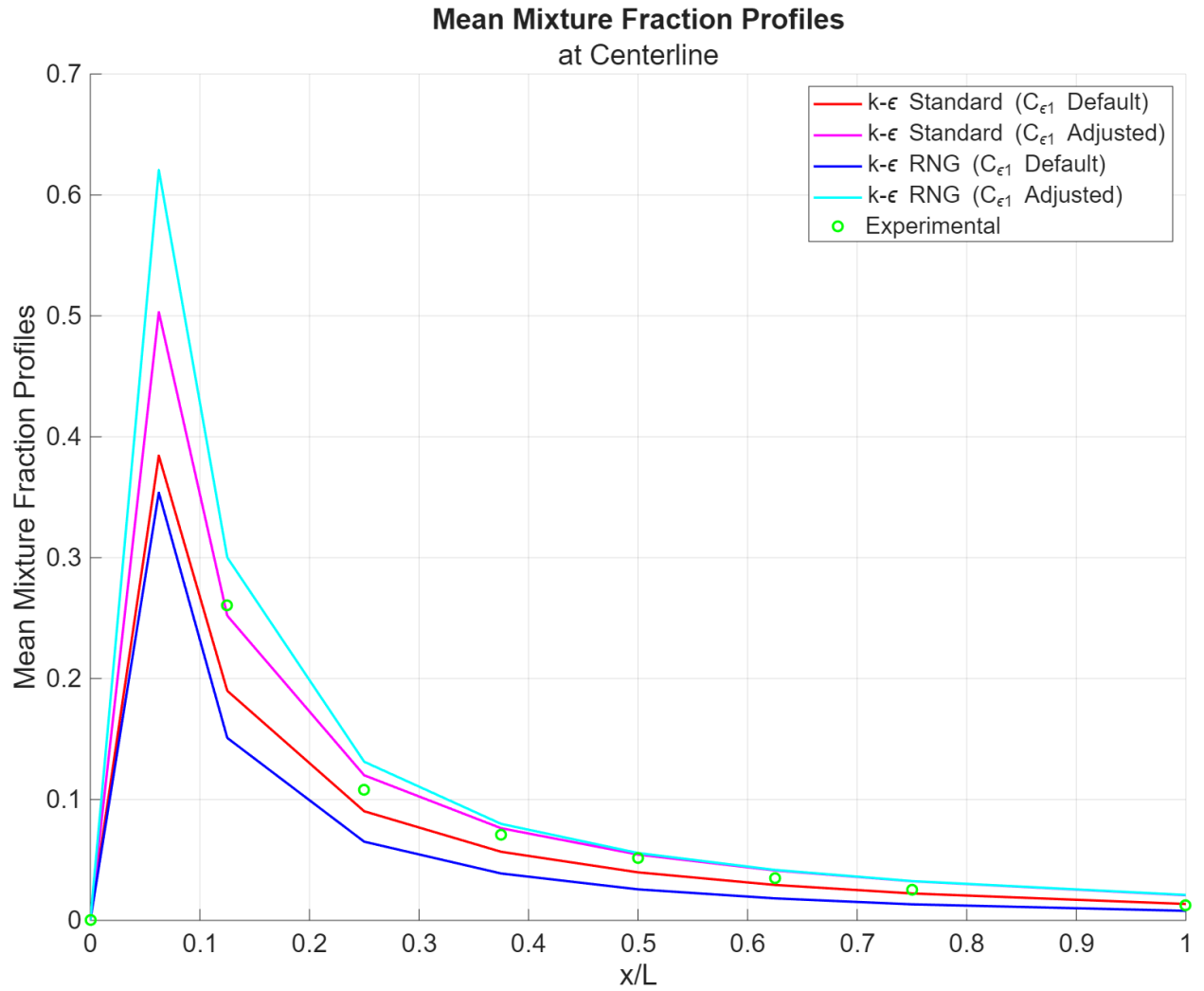


Figure 25 Mean Mixture Fraction at Centerline

Fig. 25 shows the mean mixture fraction at the centerline for each of the models and experimental data. The differing modes do not match well at the inlet with the RNG $C_{\epsilon 1}$ adjusted value predicting the highest value and the RNG $C_{\epsilon 1}$ default predicting the lowest value. The standard $C_{\epsilon 1}$ adjusted model matches the experimental data the best consistently.

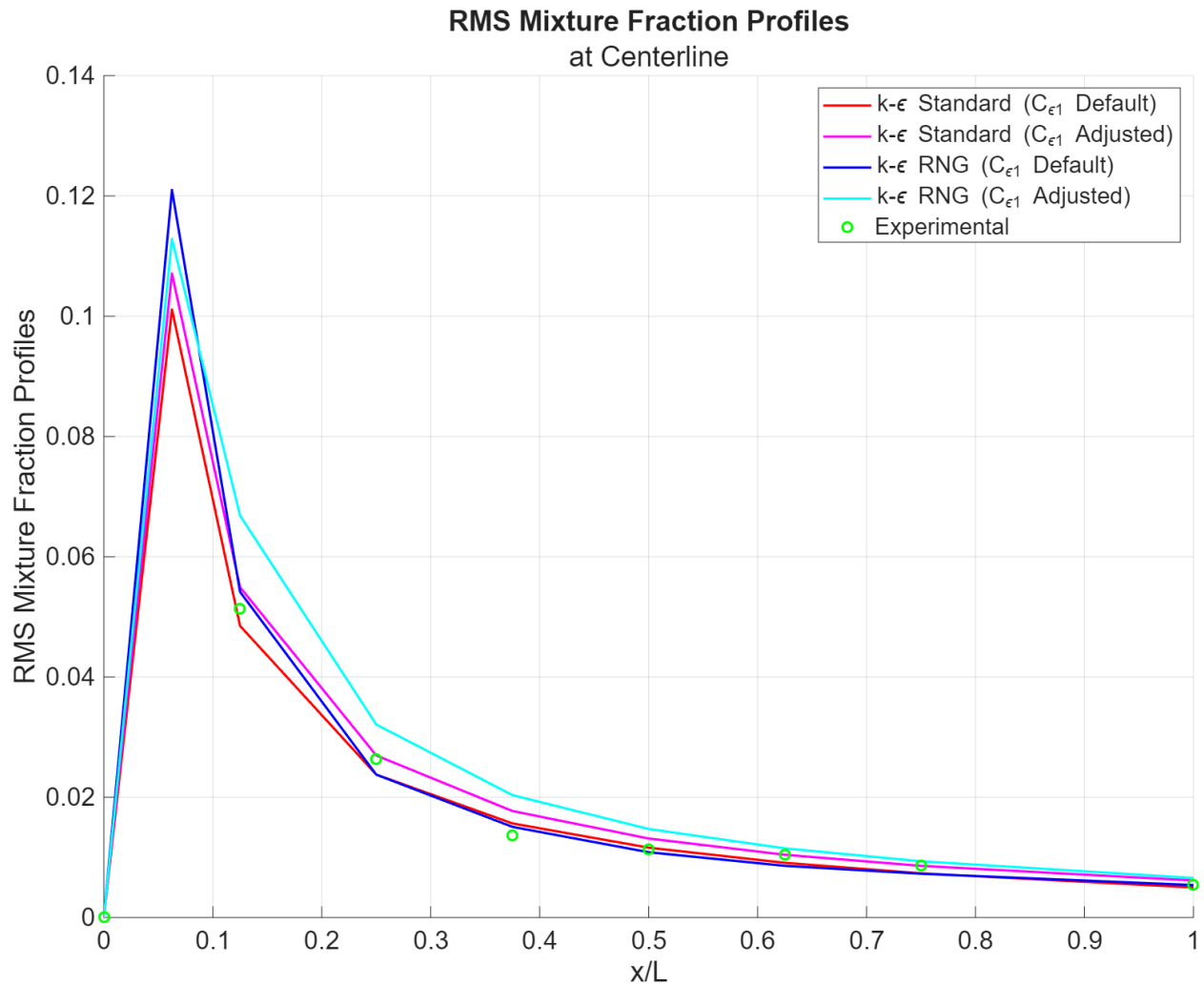


Figure 26 RMS Mixture Fraction at Centerline

Fig. 26 shows the RMS mixture fraction at the centerline for each of the models and experimental data. All the models fit well with each other and the experimental data as they trend along the same path. The standard model's $C_{\epsilon 1}$ value has less of an affect than this parameter does on the RNG model as the graphs follow better for the standard. The experimental data follows all the models well expect for the RNG $C_{\epsilon 1}$ adjusted model as it predicts a higher RMS mixture fraction than the experimental data shows.

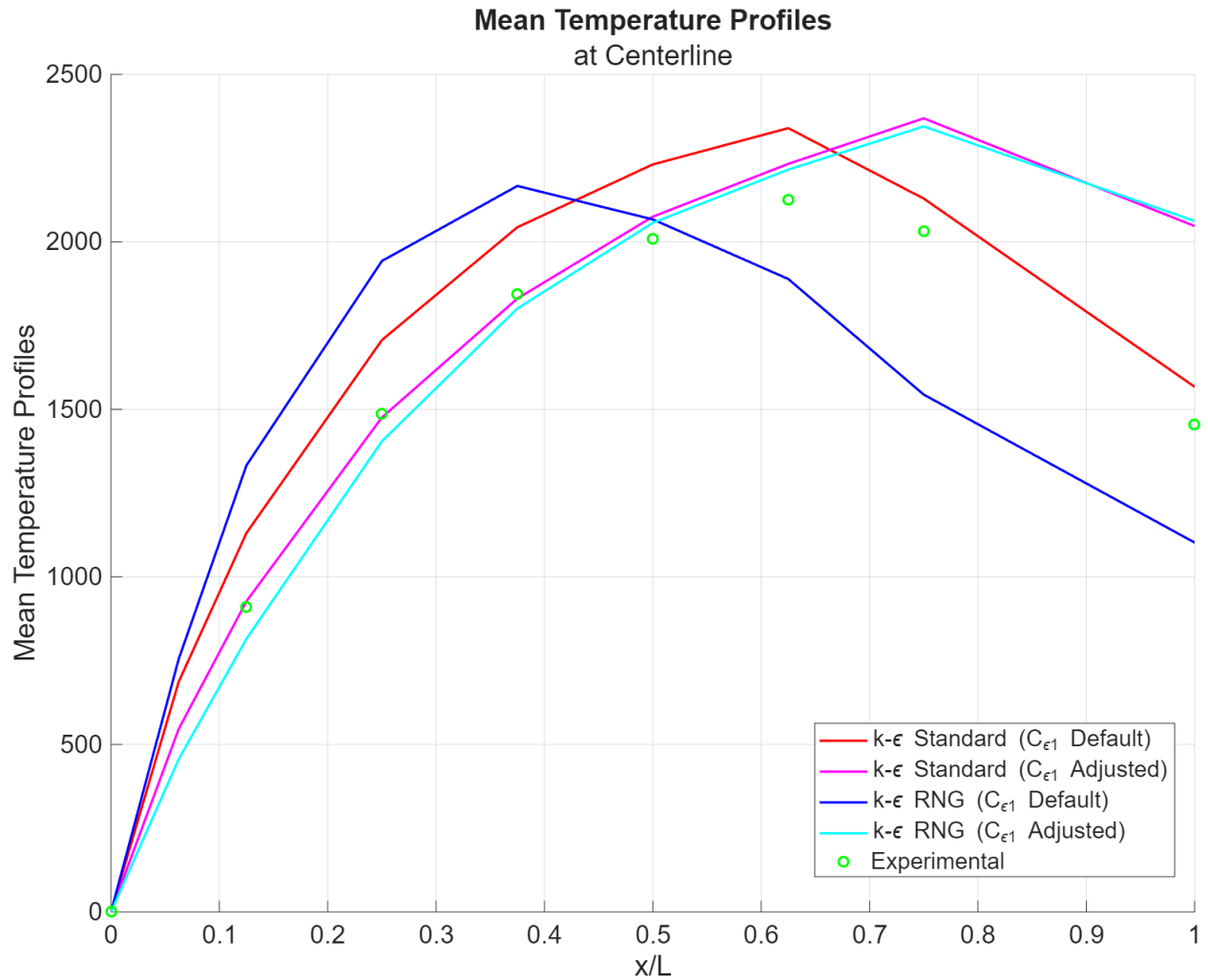


Figure 27 Mean Temperature at Centerline

Fig. 27 shows the mean temperature at the centerline for each of the models and experimental data. All the models fit well with each other and the experimental data as they trend along the same path. The $C_{\epsilon 1}$ adjusted models coincide well with each other due to the fact they are both calibrated to have similar values. The experimental data follows the standard $C_{\epsilon 1}$ adjusted model near the inlet, but the $C_{\epsilon 1}$ default value near the outlet.

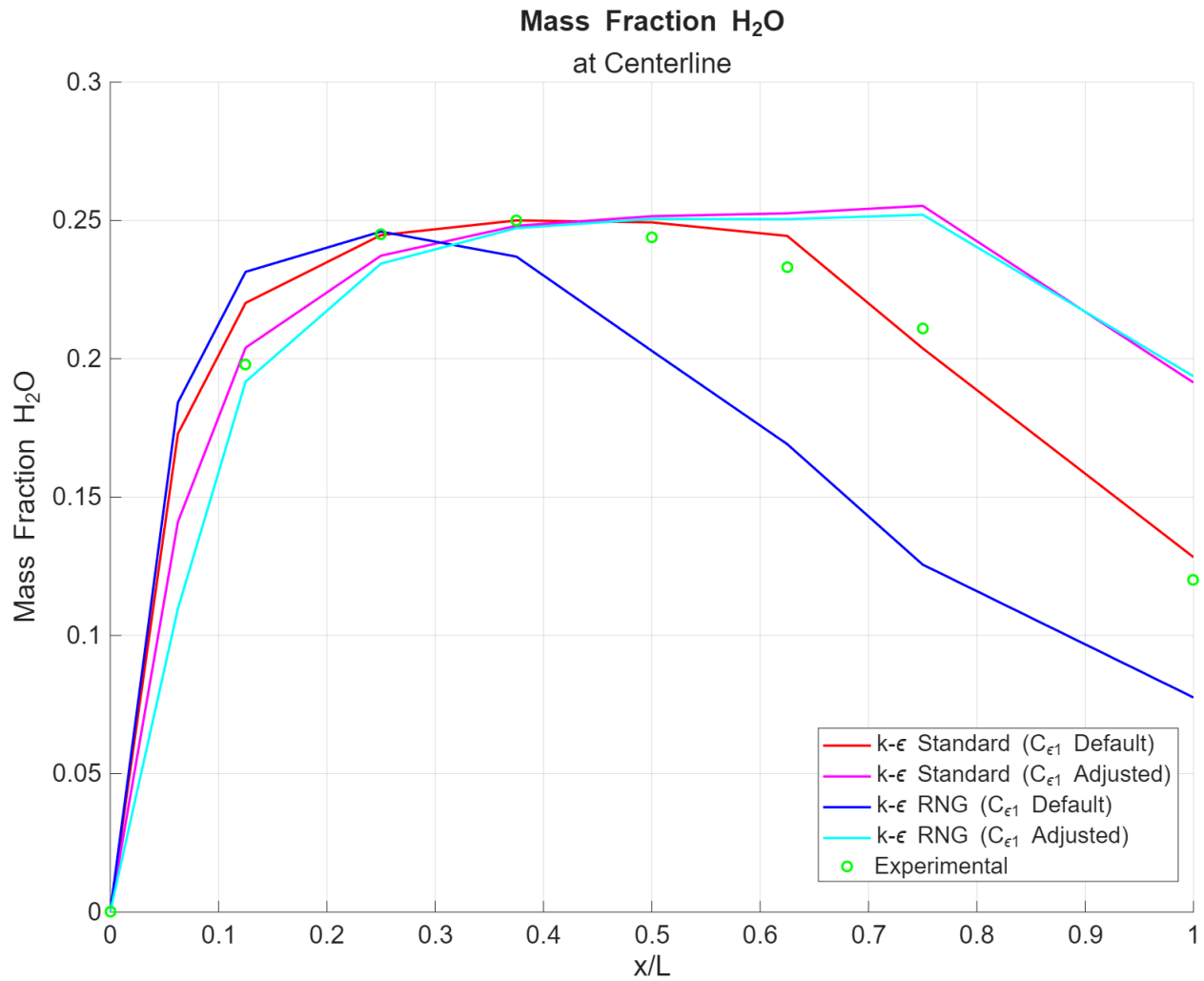


Figure 28 H₂O Mass Fraction at Centerline

Fig. 28 shows the mass fraction for H₂O at the centerline for each of the models and experimental data. All the models fit well with each other and the experimental data as they trend along the same path. The $C_{\epsilon 1}$ adjusted models coincide well with each other due to the fact they are both calibrated to have similar values. The experimental data follows the standard model with default $C_{\epsilon 1}$ the best, especially near the outlet. Near the inlet, the experimental data follows the standard $C_{\epsilon 1}$ adjusted model well.

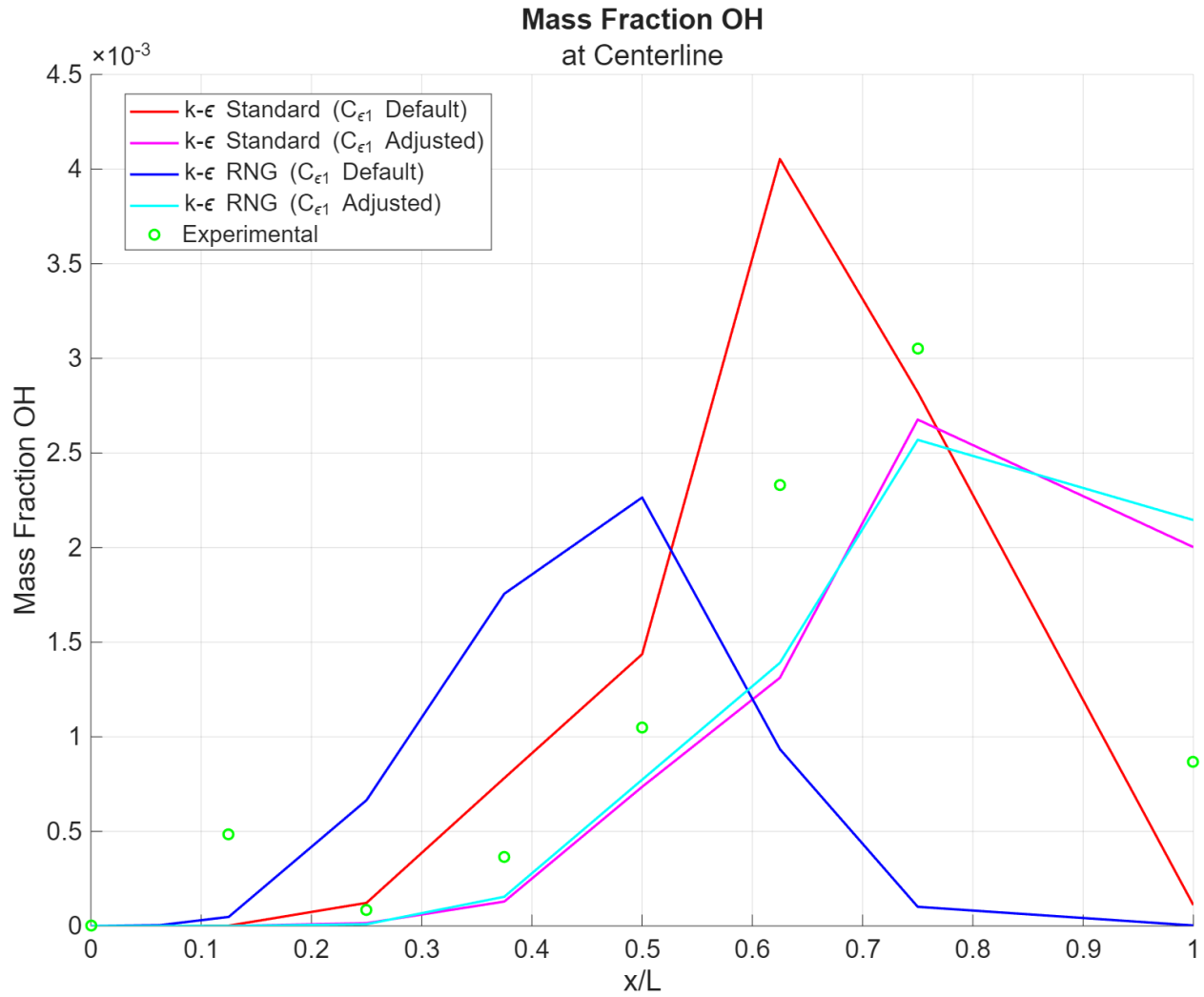


Figure 29 OH Mass Fraction at Centerline

Fig. 29 shows the OH mass fraction at the centerline for each of the models and experimental data. It is evident that the default $C_{\epsilon 1}$ values are not a good fit as the trends and values are way off. The $C_{\epsilon 1}$ adjusted models are slightly better with the trends matching up. However, the values still do not align well.

Table 5 Case Name Abbreviation and Meaning

Abbreviation	Meaning
Case 1	k- ϵ Standard ($C_{\epsilon 1}$ Default)
Case 2	k- ϵ Standard ($C_{\epsilon 1}$ Adjusted)
Case 3	k- ϵ RNG ($C_{\epsilon 1}$ Default)
Case 4	k- ϵ RNG ($C_{\epsilon 1}$ Adjusted)
Ex	Experimental

Table 5 shows the name of the cases used and their meanings. This naming system is used on Table 6 to save space.

Table 6 Maximum Percent Differences at the Centerline for Each Case and Output Value

Cases	Axial Velocity	RMS Velocity	Mean Mixture Fraction	RMS Mixture Fraction	Mean Temperature	Mass Fraction H ₂ O	Mass Fraction OH
1-2	26.8753	13.4024	42.0385	20.8091	26.5502	39.4878	191.1624
1-3	36.1922	19.0097	52.8911	17.8647	34.8177	49.2743	191.0481
2-3	60.6183	32.2069	89.9307	19.3519	59.9817	84.6447	199.6942
1-4	45.4476	21.3826	47.0254	31.8083	40.3898	44.7538	199.5175
2-4	20.2321	12.4448	20.9014	19.5641	17.9724	25.0310	179.7017
3-4	64.3348	33.5105	90.9511	30.2374	60.6485	85.5984	199.9836
1-Ex	35.7702	17.6297	31.4300	15.6247	21.7738	10.6009	198.7189
2-Ex	16.8492	16.6552	47.5896	26.2801	33.8874	45.8641	199.9510
3-Ex	63.1411	31.7387	66.9619	18.9500	37.8168	50.7401	198.8341
4-Ex	15.9719	18.3477	48.7978	39.8045	34.5995	46.9660	199.9922

Table 6 shows the maximum percent difference for each combination of model and experimental results for each output. The models tend to do a better job at predicting the experimental results for RMS velocity. They also have better consistency as the percent differences are lower with the maximum being 33%. The mass fraction for OH has the largest mass fractions with the differences being around 200% consistently. This is due to how unstable OH is which makes it hard to predict. Because Table 6 shows the maximum percent differences, it can be assumed that these are the percent differences around the edges of the flame as this is where there is the most uncertainty. The green boxes coincide with the percent differences that are less than 25 % which show models and outputs that align well. Case 2 and Case 4 align well as they are set to be close to the experimental values which allows these two cases to be calibrated. Based on the green boxes, it can be said that Case 1, k- ϵ Standard ($C_{\epsilon 1}$ Default), is the closest to the experimental values and thus the best fit for most of the output parameters. However, the centerline graphs show the k- ϵ Standard ($C_{\epsilon 1}$ Adjusted) model is a better representation of the centerline of the flame.